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Review article

Research progress on CO2 capture and utilization technology

Lipei Fu [a,b,c], Zhangkun Ren [b], Wenzhe Si [a], Qianli Ma [b], Weiqiu Huang [b], Kaili Liao [b], Zhoulan Huang [a], Yu Wang [d], Junhua Li [a], Peng Xu [c]

[a] National Engineering Research Center for Synergetic Control of Air Pollutants and Greenhouse Gases, School of Environment, Tsinghua University, Beijing 100084, PR China
[b] School of Petroleum Engineering, Changzhou University, Changzhou 213164, PR China
[c] Jiangsu NanFang Bearing Co.,Ltd., Changzhou 213164, PR China
[d] School of Environmental Science and Engineering, Yancheng Institute of Technology, Yancheng 224051, PR China

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ABSTRACT

With the expansion of industry, the emission of greenhouse gases is increasing, and its impact on climate is becoming more and more serious. CO_2 is the main culprit of the greenhouse effect, and how to effectively solve the climate problem caused by CO_2 has attracted more and more attention. In recent years, there have been continuous attempts to reduce CO_2 emissions from the source, but no obvious results have been achieved. In fact, CO_2 is not only a greenhouse gas, but also a potential carbon resource. Therefore, how to capture and effectively use CO_2 is also the research direction that many scholars have been exploring recently. In this paper, the current situations of CO_2 capture technologies are reviewed from the aspects of chemical absorption, solid-phase porous materials adsorption, membrane separation, cryogenic separation, hydrate method and microbiological method in the first part. Then, the CO_2 utilization technologies are systematically introduced from the aspect of physical utilization, chemical utilization, biological utilization and mineralization utilization. Furthermore, several representative frontier technologies of CO_2 resource utilization are reported. On this basis, the advantages and disadvantages of different methods are summarized to provide some ideas and references for alleviating the CO_2 issue.

1. Introduction

Since the start of the industrial revolution, global greenhouse gas emissions have been increasing, and the greenhouse effect has become more and more serious. Climate change has become a serious challenge to be faced. According to statistics, CO_2 is the main gas causing the greenhouse effect, accounting for more than 70% of the global greenhouse gas, and the rest are methane, nitrous oxide, and various fluorinated gases [1]. Since 1970, CO_2 emissions have increased by about 90%, from 16GT (16×10^9 t) in 1970–37.5 GT (37.5×10^9 t) in 2018 [2,3]. At the same time, the increase of CO_2 emissions intensifies the greenhouse effect, leading to a series of problems, such as sea-level rise, glacier melting, and ocean acidification, which has aroused human concern about the change in the ecological environment [4]. In this context, how to effectively solve the problem of CO_2 emissions has become a hot research topic.

Carbon capture and storage (CCS) technology refers to the process of separating CO_2 from relevant emission sources, then transporting it to the storage site and isolating CO_2 from the atmosphere for a long time. This technology can reduce CO_2 emissions from the source, and can remove the existing CO_2 in the atmosphere on a large scale, which is a relatively good method to alleviate the greenhouse effect [5]. However, from the current situation, the high cost of carbon capture (20–120 $/t) and the leakage risk of geological storage seem to be the bottlenecks hindering the large-scale development of CCS [6]. In this context, a wide based strategy is needed to ensure long-term stability [7]. This has prompted scholars to shift their research focus to carbon capture, utilization and storage (CCUS) technologies [7,8].

CCUS is an emerging CO_2 emission reduction technology with great potential, and it is also a new development trend of CCS technology. The CO_2 captured in the production process is put into a new production process and can be converted and utilized in different ways. This process is not simply storage, but effectively realizes the recycling and reuse of CO_2 (Fig. 1). In contrast, CCUS technology can recycle CO_2 and generate economic benefits, which is more practical [9]. At present, the concept of CCUS has been gradually accepted worldwide, and the related CO_2

E-mail addresses: siwenzhe115@tsinghua.edu.cn (W. Si), lijunhua@tsinghua.edu.cn (J. Li).

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capture and utilization technologies have also been widely explored. This paper mainly reviews the research status of various CO_2 capture technologies (absorption, adsorption, membrane separation, cryogenic separation, hydrate technology, and microbial technology) and utilization technologies (physical utilization, chemical utilization, biological utilization, and mineralization utilization), as shown in Fig. 1. Furthermore, several representative frontier technologies of CO_2 resource utilization are reported. In addition, the advantages and disadvantages of different CO_2 capture and utilization technologies are systematically analyzed, so as to provide some ideas for mitigating the climate change trend.

2. CO2 capture

In general, CO_2 capture technology routes can be divided into three types: pre-combustion capture, post-combustion capture, and oxy-fuel combustion technology [1] (Fig. 2, Table 1). Because pre-combustion capture and oxy-fuel combustion capture technologies require appropriate materials and certain conditions to meet high-temperature requirements, the application research and development of these two technologies are relatively few. In contrast, post-combustion capture is a widely used and mature technology in the industry, with good CO_2 selectivity and capture efficiency [10]. Various post-combustion capture techniques, including chemical absorption, adsorption, membrane, cryogenics, hydrate, and microbial are explored in this review, with their challenges and future prospects identified.

2.1. Chemical absorption

Chemical absorption is a method that uses chemical solvents (usually weak alkaline solution in industry) to react with CO_2 to generate compounds, and then realizes the desorption of CO_2 by changing external conditions (pressure and temperature), so that the absorbent can be recycled [11]. At present, the absorbents studied mainly include organic amine solution, ammonia solution, and sodium hydroxide solution. Table 2 summarizes the advantages and disadvantages of several common chemical absorbents. Among them, the capture of CO_2 by organic amine solution is the most widely used and effective method in chemical absorption [12].

2.1.1. Organic amine solution

The capture of CO_2 by organic amine solution appeared in the 1930 s and has been widely developed and applied in CO_2 capture due to its good performance and lower toxicity [13]. After decades of



Fig. 1. Schematic diagram of CO_2 capture and utilization cycle.

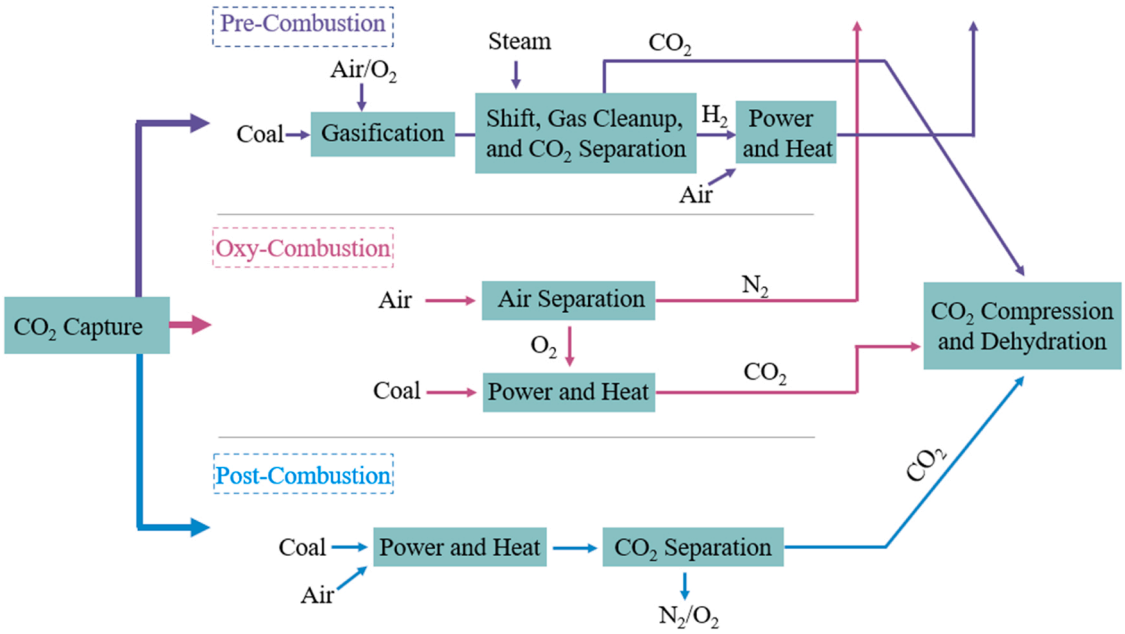


Fig. 2. Roadmap of different carbon capture technologies for power plants.

Table 1
Summary of advantages and disadvantages of CO_2 capture routes [1,10].

Capture method	Advantages	Disadvantages
Pre-combustion capture	Less regenerative energy; small capture equipment required	High investment cost; operation is limited
Oxy-fuel combustion capture	High concentration of captured CO_2; less NO_x formation	High oxygen demand increases investment costs and energy consumption
Post-combustion capture	Low investment cost; fast technology introduction; flexible operation to reduce operating costs	Lower CO_2 partial pressure in flue gas increases capture energy consumption and cost, large capture equipment required

Table 2
Advantages and disadvantages of different chemical absorbents [1,12,17,21].

Absorbent	Advantages	Disadvantages
Organic amine solution	Mature technology; strong absorptive capacity	High energy consumption; be corrosive
Ammonia	Low corrosivity; low cost; lower energy consumption than organic amine solution	High escape of ammonia; large equipment required
Sodium hydroxide solution	Low cost; high absorption efficiency	Strong corrosive; high energy consumption for absorbent regeneration

development, the technology of organic amine solution to capture CO_2 has become more and more proficient. Due to its excellent capture capacity and separation efficiency, scholars have carried out a large number of studies on the absorption of CO_2 by organic amine solution.

Park et al. [14] proposed a new method to remove CO_2 by combining CCS and the CO_2 fixation process. The monoethanolamine (MEA), diethanolamine (DEA), and methyldiethanolamine (MDEA) were used to make 10 wt% or 30 wt% amine solution, and then the three amine solutions were respectively added to the saturated CO_2 solution containing Ca^{2+} for CO_2 fixation experiment. The results showed that all three amine solutions could convert more than 84% (mass fraction) of CO_2 into precipitates, and the carbonate formation rate of tertiary amine (MDEA) was higher than that of MEA and DEA. When the product carbonate was analyzed by X-ray diffraction (XRD), it was found that its composition depended on the type of amine solution. It followed that CO_2 could be fixed with an amine solution and Ca^{2+}. In addition to using Ca^{2+}, Park et al. [15] also used $BaCl_2$ as a metal ion source mixed with an amine solution for CO_2 fixation. The results demonstrated that the Ba^{2+}-containing amine solution reacted with aqueous CO_2 to form precipitates rapidly. It was proved that barium ion has a CO_2 fixation potential similar to calcium ion.

Although the organic amine solution has an outstanding ability to absorb CO_2, its desorption process needs to consume a lot of energy, and the desorption energy consumption accounts for about 70%− 80% of the total energy consumption for CO_2 capture. Therefore, solving the energy consumption problem of desorption of organic amine solution is a crucial step in the development of CO_2 capture technology in the future. Yin et al. [16] tested the absorption and desorption properties of four monoethanolamine-based (MEA/water, MEA/ethanol, and two kinds of MEA/ethanol/ water) absorbents. The results indicated that the MEA added with ethanol had better desorption effect, lower desorption temperature, and good regeneration ability. Compared with the MEA/water system, the desorption temperature of the MEA/ethanol system was decreased by 20 °C, the average desorption rate in the first 20 min was increased by 41.72%, and the regeneration rate increased by 5.52%. In contrast, the desorption performance of the MEA/ethanol/-water system was extremely unsatisfactory. It was worth noting that under the condition of 1 mol MEA loaded with 0.07 mol CO_2, the desorption energy of the MEA/ethanol system was 80% lower than that of the MEA/water system, and the lower CO_2 loading was beneficial to the absorption of CO_2 by the absorbent. Therefore, the addition of organic solvents such as ethanol is favorable to reduce the desorption energy consumption and desorption temperature of the amine solution and promote the absorption of CO_2 by the amine solution.

2.1.2. Ammonia

The process of absorbing CO_2 by aqueous ammonia is somewhat similar to that of organic amine solution. However, due to the low average efficiency of aqueous ammonia absorbing CO_2 and the high escape amount of ammonia, the former is not as widely used as the latter. The reaction process of CO_2 capture by aqueous ammonia is complex, and the overall reaction formula is: NH_3 +CO_2

$+H_2O \leftrightarrows 2NH_4HCO_3$. Specifically, it is divided into the following reactions [17]:

$2NH_3+CO_2=NH_2COONH_4$

$NH2COONH4 + H2O \leftrightarrows NH4HCO3 + NH3$

$NH4HCO3 + NH3 \leftrightarrows (NH4)2CO3$

$(NH4)2CO3 + H2O + CO2 \leftrightarrows 2NH4HCO3$

The advantages of CO_2 capture by aqueous ammonia are low energy consumption, low corrosivity and low cost. However, the escape of ammonia has always been a big problem for this technology. In general, as a new technology, ammonia capture of CO_2 still has a broad application prospect.

Liu et al. [18] proposed a regenerative ammonia method to recycle the absorbent by removing CO_2. By combining the two processes of normal temperature absorption and heating regeneration, the regeneration and utilization of ammonia were realized. The experimental results showed that the CO_2 capture efficiency of the regenerative ammonia method could reach more than 90%. Compared with MEA method, the energy consumption of the regenerative ammonia method is relatively low, and it has certain advantages in future development. However, it is still necessary to further improve the escape of ammonia. Ma et al. [19] studied the inhibitory effect of polyethylene glycol dimethyl ether (NHD) on ammonia escape. The results demonstrated that the addition of a certain amount of NHD solution could effectively inhibit the escape of ammonia, which promoted the decarbonization efficiency of aqueous ammonia to a certain extent. However, different concentrations of NHD had certain adaptability to different concentrations of aqueous ammonia. In this experiment, adding 5% NHD reduced the escape amount of ammonia by 24.86% and increased the CO_2 removal efficiency by 10%.

Based on the traditional ammonia method for CO_2 capture, Siddiqui et al. [20] investigated a new wind energy-based CO_2 capture system (Fig. 3). It used wind turbines to produce and use synthetic ammonia to capture CO_2 on site. The captured CO_2 was used to produce ammonium bicarbonate, which could be used in other fields. Such as the production of food products or applications in plastic and rubber industries. In this study, the CO_2 capture cost was about 0.1–0.23 \$/kg CO_2, the CO_2 capture rate was 3.5 kg/s, and the energy consumption was found to be 640.1 kg CO_2/MWh. Therefore, the capture system has certain development potential.

2.1.3. Sodium hydroxide solution

NaOH solution has strong CO_2 absorption capacity and low cost, and gradually shows its advantages in recent years. The mechanism of CO_2 absorption by sodium hydroxide is that NaOH is a strong base, and its aqueous solution is facile to react with CO_2. The reaction formula is as follows [21]:

$2NaOH + CO2 \leftrightarrows Na2CO3 + H2O$

Continue to pass in CO_2:

$Na2CO3 + H2O + CO2 \leftrightarrows 2NaHCO3$

It can be seen from the reaction formula that carbonate is generated after absorption. If the absorbent is regenerated and CO_2 is separated from the carbonate, the capture cost will be greatly increased. Therefore, alleviating the high energy demand of absorbent regeneration is the key

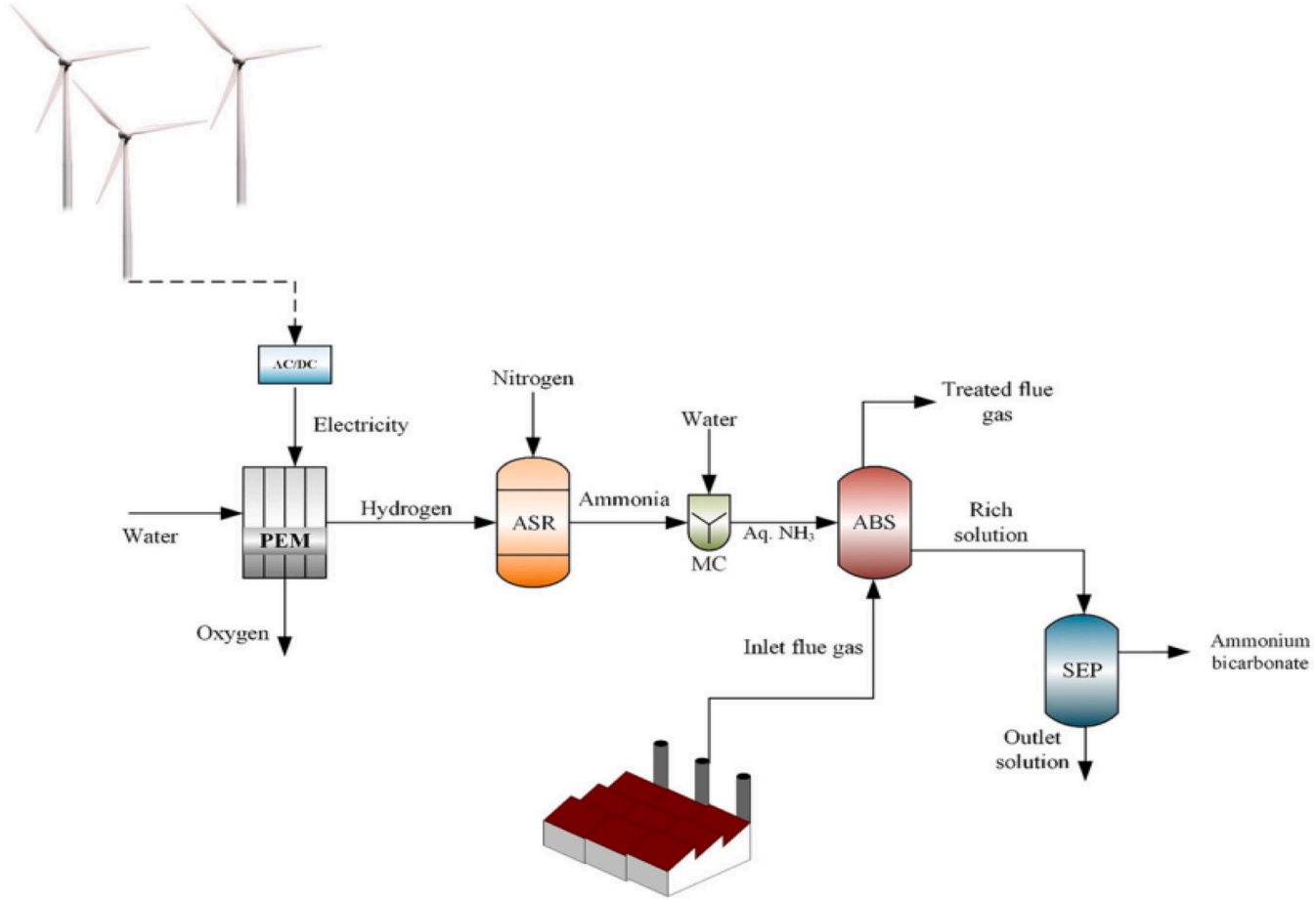


Fig. 3. New wind energy CO_2 capture system with ammonia absorbent [20].

to the capture of CO_2 by sodium hydroxide. Mahmoudkhani et al. [22] developed a new method for separating NaOH from basic sodium carbonate solution: first, the concentrated NaOH solution was precipitated and crystallized to separate anhydrous sodium carbonate, and then the anhydrous sodium carbonate was causticized with sodium titanate. During the causticization process, sodium hydroxide was regenerated and CO_2 was separated into pure gas stream, as shown in Fig. 4. The experiment presented end-to-end energy and exergy analysis, and the results showed that this method reduced the energy consumption by 50% compared with the traditional causticizing technology, and the CO_2 capture cost was greatly reduced. In addition to the disadvantage of high energy demand for absorbent regeneration, the low solubility of CO_2 in water made its absorption efficiency in NaOH solution lower, which also limited the large-scale application of NaOH for CO_2 capture. Chiang et al. [23] found that adding glycerin, a by-product of biodiesel waste, into NaOH aqueous solution could improve the CO_2 absorption efficiency of NaOH solution. In a rotating packed bed (RPB), the addition of glycerol to NaOH solution could easily increase the CO_2 absorption efficiency to more than 90%. Under the same operation, adding 7 wt% glycerol to 0.5 mol/L NaOH solution could make its CO_2 absorption efficiency reach 95%, while the absorption efficiency of glycerol free NaOH solution was only 70%. At high rotating speed, the chemical enhancement factor could reach more than 50 with the addition of glycerol. The experimental results indicated that the addition of glycerol made the NaOH solution possessed a better CO_2 capture effect.

2.2. Solid-phase porous materials adsorption

The solid material adsorption method utilizes the van der Waals force or electrostatic force between the porous solid (adsorbent) and CO_2 (adsorbate) to separate and capture CO_2 [24]. The adsorption separation method is regarded as a potential CO_2 capture method due to its high adsorption capacity, low regeneration cost, no corrosion, and low energy consumption [25,26]. The ideal adsorbent should have high adsorption capacity and selectivity for CO_2, and be easy to regenerate for economic feasibility [27,28]. At present, the most studied adsorption materials include activated carbon, zeolite, Metal organic frameworks (MOFs), Covalent organic frameworks (COFs), M41s series mesoporous materials (MCM-41, MCM-48 and MCM-50) and hydrotalcites, etc. [29,30]. In addition, geopolymers, as emerging adsorption materials in recent years, have also shown great potential in CO_2 capture [31]. Table 3 lists the significant advantages and disadvantages of several solid materials for CO_2 adsorption. Although it is effective and feasible for solid material to adsorb CO_2, the amount of CO_2 adsorption is much smaller than that of absorption methods. Therefore, many scholars have been working hard to improve the adsorption capacity of CO_2 through the modification of solid materials.

2.2.1. Activated carbon

Activated carbon is a common physical adsorbent with abundant microporous structure and large specific surface area. The specific surface area of ordinary activated carbon is between 500 and 1700 m^2/g. The advantages of activated carbon mainly include strong adsorption capacity, low production cost, good chemical stability, easy regeneration, etc. So it is widely used in industrial production, environmental restoration, energy storage, etc., which is a kind of adsorbent with great development prospects [26,32]. At present, many researches have focused on modified activated carbon. By modifying the activated carbon, the adsorption capacity of the activated carbon is further improved, achieving some good results.

Table 3
Advantages and disadvantages of CO_2 solid adsorption materials [1,26,29,32,45,52,59].

Adsorbent material	Advantages	Disadvantages
Activated carbon	High stability; high specific surface area; low cost of regeneration	Poor water resistance; adsorption capacity is greatly affected by CO_2 partial pressure
Zeolite	Long adsorption period; high adsorption capacity; aperture uniform	Low gas selectivity; adsorption capacity is greatly affected by temperature and water vapor
MOFs	Tunable pore structure; high adsorption capacity at high pressure	High cost; poor adsorption capacity at low pressure; material synthesis is complex
COFs	Lightweight material; adjustable structure; high stability	High cost; material synthesis is complex
M41s series mesoporous materials	High specific surface area; well-developed pore structure; aperture size adjustable	Low adsorption capacity; adsorption capacity is greatly affected by water vapor
Hydrotalcites	Functional under a wide temperature and pressure range	Low adsorption capacity
Geopolymers	Low cost of synthesis	Low adsorption capacity

Sarwar et al. [33] prepared hydrochar (HC) by hydrothermal carbonization (HTC), and then thermochemically activated HC with H_3PO_4, $ZnCl_2$ and KOH to generate porous activated carbon samples (Fig. 5a). These active substances led to the surface modification of the activated carbon, which not only increased the specific surface area and pore volume of the activated carbon, but also enhanced the adsorption capacity of the activated carbon for CO_2. At 15 °C and 1 bar pressure, the KOH-modified activated carbon expressed the highest CO_2 adsorption capacity (4.5 mmol/g). This result is higher than most experimental data under the same conditions. Xu et al. [34] also thermally activated the urea-modified carbon spheres with KOH to obtain highly porous nitrogen-doped activated carbons (Fig. 5b). The results showed that the adsorption capacity of the modified adsorbent for CO_2 reached 4.37 mmol/g at 25 ℃. On this basis, Shi et al. [35] found that the nitrogen-doped activated carbon obtained by KOH microwave activation had greater CO_2 adsorption capacity. The experimental results indicated that the adsorption amounts of CO_2 of the two nitrogen-doped activated carbons prepared by KOH thermal activation were 3.44 mmol/g and 3.09 mmol/g respectively at 25 ℃, while the CO_2 adsorption capacity after KOH microwave activation was increased to 4.21 mmol/g and 3.57 mmol/g respectively.

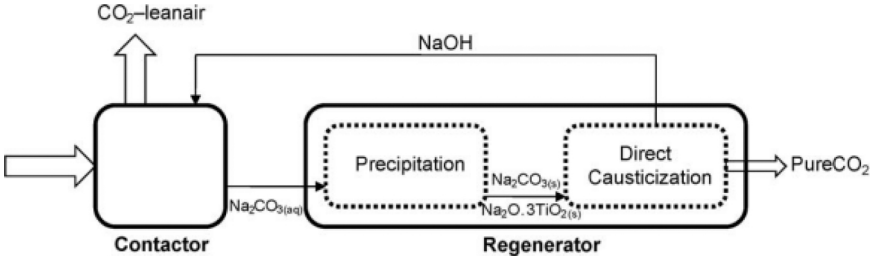


Fig. 4. Schematic diagram of NaOH recovery [22].

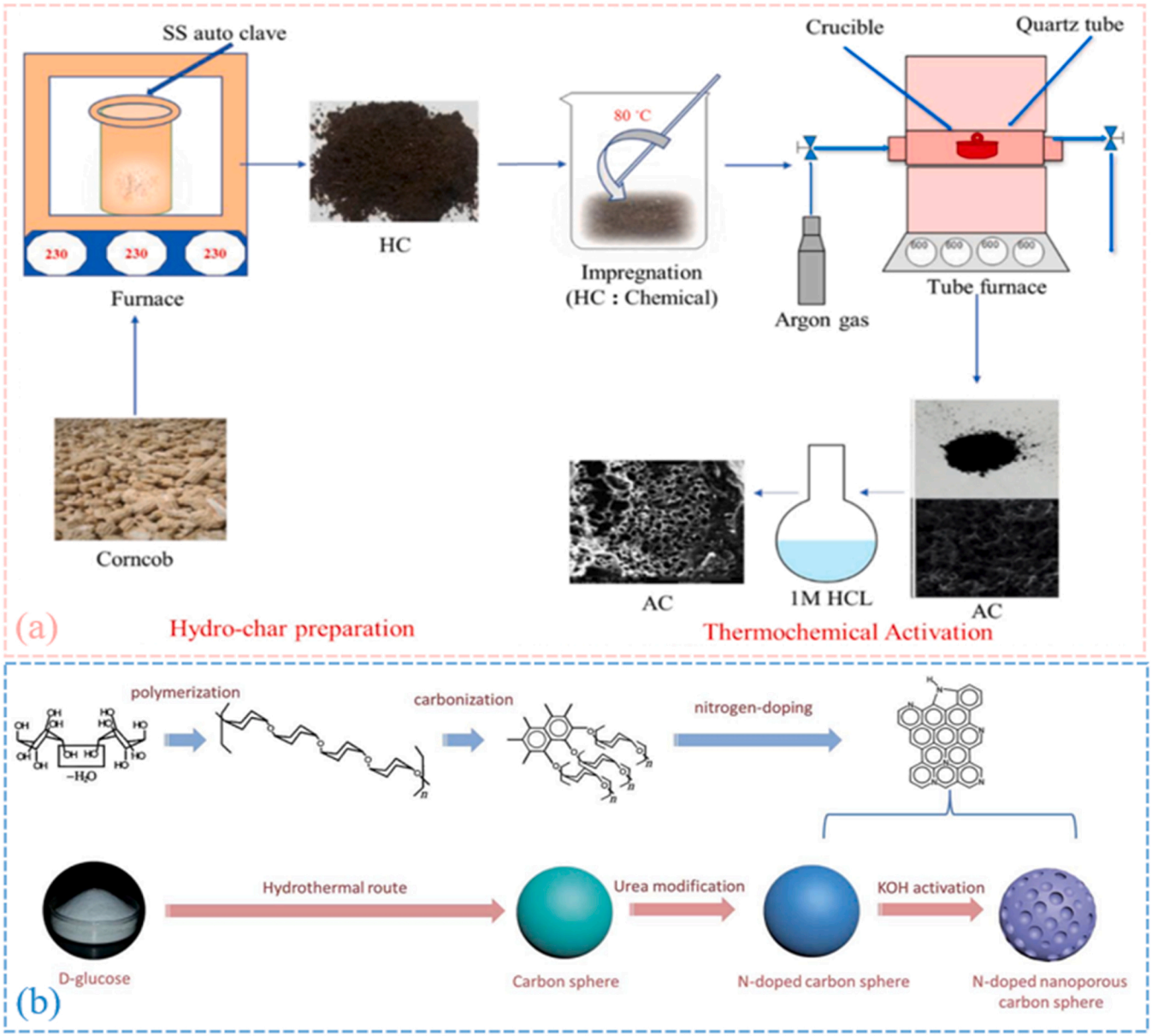


Fig. 5. The thermal activation modification process of two activated carbons [33,34].

2.2.2. Zeolite

In studies on CO_2 adsorption and separation, zeolite has become an important adsorbent material owing to its unique structure and cation exchange potential [36]. The adsorption capacity of zeolite is very large. The adsorption capacity of natural zeolite for CO_2 is generally 0.9–1.4 mmol/g. After adsorption, it can be activated and recycled repeatedly by changing the pressure. Compared with activated carbon, zeolite has a long service life, and its adsorption effect is greater than that of activated carbon, so it does not need to be replaced repeatedly [37]. The disadvantage of zeolite is that its adsorption capacity is greatly affected by temperature and CO_2 partial pressure [38]. In addition, zeolite is generally only suitable for anhydrous environments since they are particularly sensitive to water vapor, which results in reduced CO_2 selectivity [29].

Kim et al. [39] coated the zeolite 13X with octadecyltrimethoxysilane (ODTMS) by adding hydrophobic coating, which improved the moisture resistance and CO_2 selectivity of the zeolite (Fig. 6a). Under the indoor simulation conditions, the adsorption capacity of modified zeolite for CO_2 increased by 33.3%, and that for water decreased by 41.6%. This method provides a direction for indoor CO_2 adsorption. Kim et al. [40] and Kwon et al. [41] grafted ethylenediamine and benzylamine onto the micropore walls of HY zeolite and β zeolite by amine functionalization respectively, which also improved the hydrophobicity of zeolite. The ethylenediamine functionalized HY zeolite is exhibited in Fig. 6b. Under the condition of wet flue gas, both modified zeolites showed greater CO_2 adsorption capacities than pristine zeolites. This strategy is expected to make zeolite a powerful CO_2 adsorbent for wet flue gas.

2.2.3. MOFs

MOFs are crystalline porous frameworks assembled from organic and inorganic building blocks with strong adsorption properties [42]. Since MOFs can be composed of organic linking ligands and different transition metals (metal ions or metal clusters) through coordination bonds, different MOFs can be designed through directional adjustment (Fig. 7) [43,44]. Compared with traditional porous materials such as activated carbon and zeolites, this remarkable and easy-to-design property of MOFs is the key to exhibit their adsorption advantages [45]. MOFs are

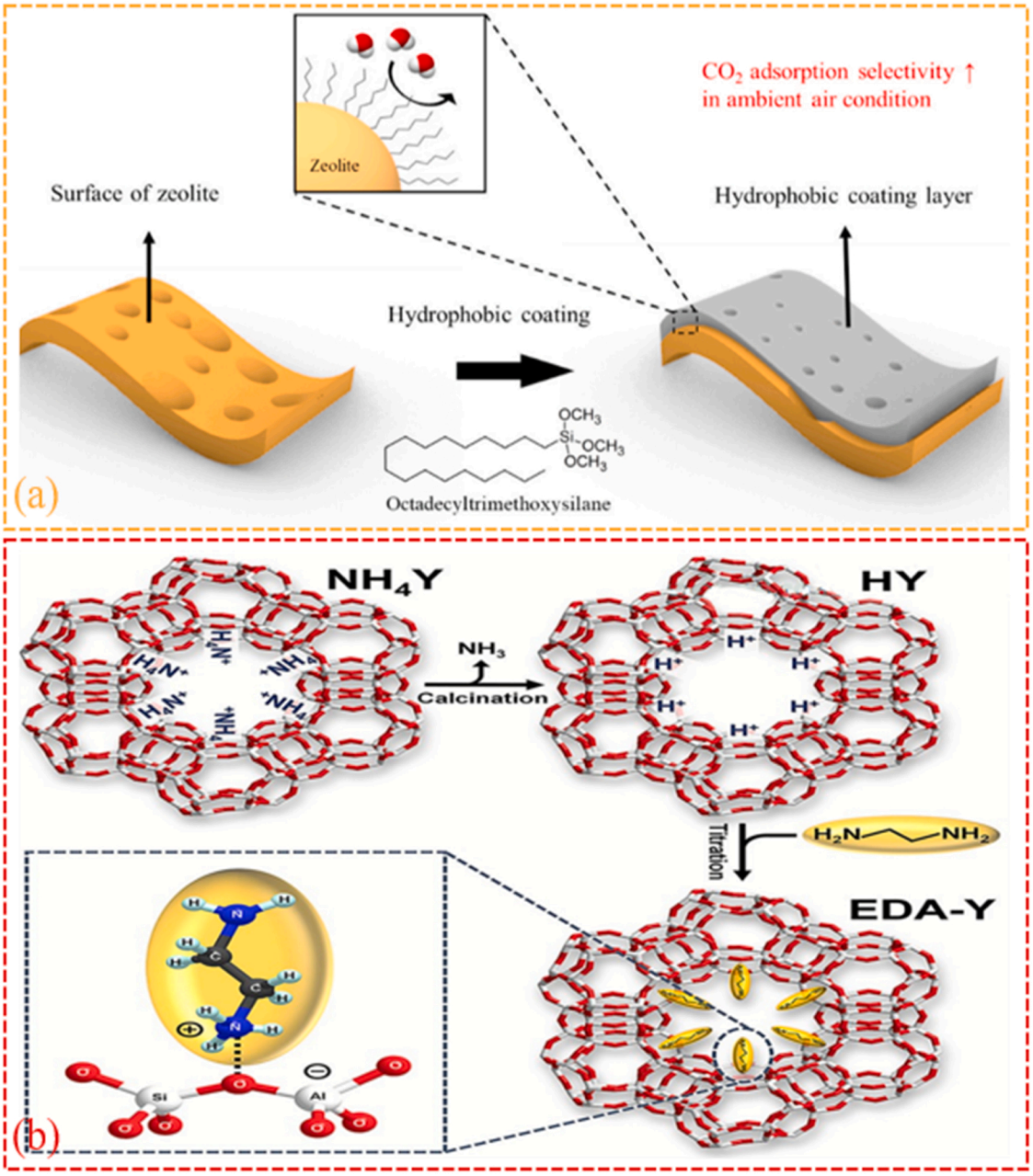


Fig. 6. (a) ODTMS hydrophobically modified zeolite 13X (b) Ethylenediamine functionalized HY zeolite [39,41].

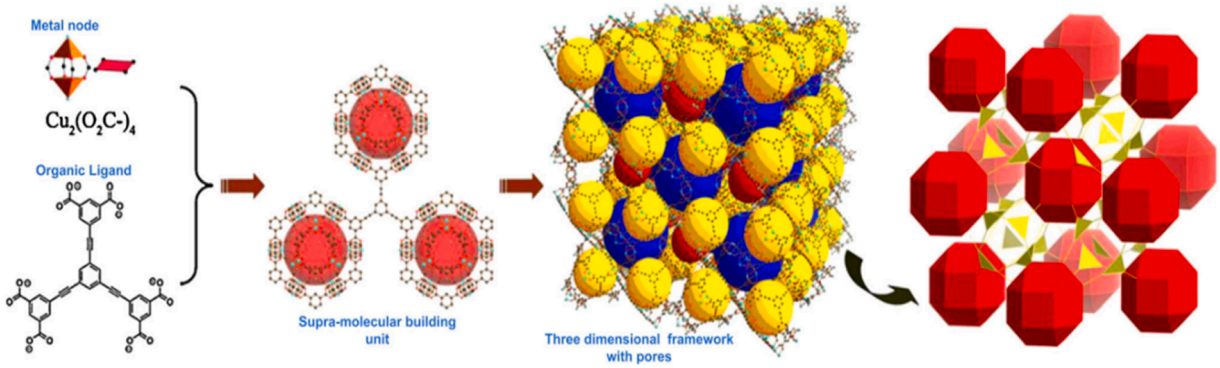


Fig. 7. Design and construction process of MOFs materials [44].

used for CO_2 capture, which can not only effectively reduce CO_2 concentration, but also convert it into valuable items [43]. However, MOFs are unstable, and various factors such as pore volume, characteristics of the pore, metal sites, and pressure will limit the adsorption capacity of CO_2 [46]. Therefore, CO_2 adsorption by MOFs is still in the research stage.

Lei et al. [47] synthesized high-performance Ni-based-MOF material MOF-74(Ni) by a facile condensation reflux approach (Fig. 8). The structure of this MOFs can be adjusted according to different synthesis conditions. MOF-74(Ni) synthesized at 140 ℃ for 24 h had an adsorption capacity of 8.29 mmol/g of CO_2 at standard temperature and pressure, which was much higher than the adsorption materials reported above. Under similar conditions, the CO_2 adsorption capacity of MOF-74 (Ni) synthesized by this method was about twice that of MOF-74(Ni) previously reported. Tapiador et al. [48] synthesized a novel Zn-based-MOF material, named Zn-URJC-8, containing two organic ligands (2-aminoterephthalic acid and 4,4-bipyridyl). The CO_2 adsorption capacity of this material was 1.28 mmol/g at 25 ℃. In addition, this MOF was used as a catalyst for CO_2 cycloaddition for the first time. Under certain conditions, the conversion of epoxide was almost 100%. This new material not only exhibits good CO_2 adsorption performance, but also is a potential CO_2 conversion catalyst.

In addition to the synthesis of new MOFs, the modification of MOFs is also one of the important means to improve the CO_2 adsorption capacity. Gaikwad et al. [49] and Jun et al. [50] used tetraethylenepentamine (TEPA) to modify MOF-177 and MOF-808 with amine groups, which improved the adsorption performance of MOFs under low pressure conditions. The CO_2 adsorption capacity of the two modified MOFs increased by 4.8 times and 7 times, respectively, under low pressure conditions. Amino-based modification of MOFs may be a new way to improve CO_2 adsorption capacity, which needs to be investigated by more scholars.

2.2.4. COFs

COFs are a kind of porous crystalline materials, which possess framework with extended structure connected by light elements (H, C, N, O, B) through covalent bond. It has the characteristics of large specific surface area, high porosity, high crystallinity, adjustable structure and so on [51–53]. Fig. 9 shows two typical two-dimensional and three-dimensional COFs. Compared with MOFs, COFs have lower density, lighter constituent elements (H, C, N, O, B), adjustable pore size and structure, and better thermal stability. Their properties remain stable at 500–600 °C [29,53,54]. Therefore, COFs are considered as potential CO_2 adsorbents.

Lan et al. [56] found that the doping of metal Li, Sc and Ti could significantly improve the CO_2 adsorption capacity of COFs, but the combination of Sc and Ti with CO_2 molecules made the desorption process difficult. Under the condition of 298 K and 1 bar pressure, the CO_2 adsorption capacity of Li-doped COF-105 and COF-102 were 7.8 and 9.3 mmol/g, respectively, which were 4 times and 8 times that of

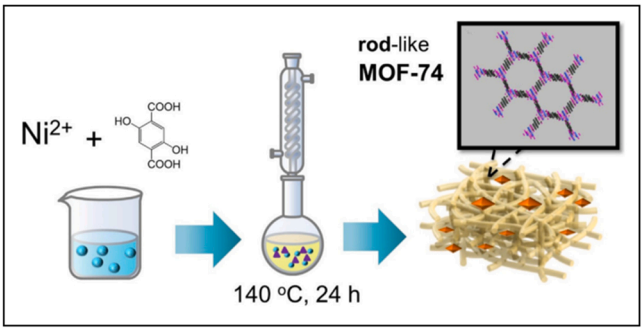


Fig. 8. Schematic diagram of synthesis of MOF-74(Ni) by condensation reflux method [47].

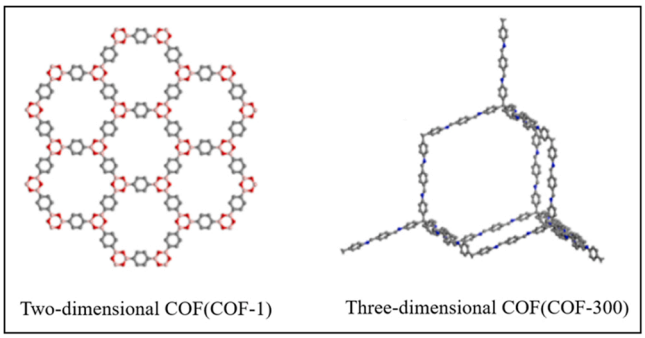


Fig. 9. Schematic diagram of typical 2D COF and 3D COF structure [55].

undoped COF-102. When the pressure increased to 40 bar, the CO_2 adsorption of Li-doped COF-102 and COF-105 reached 30.7 and 51.5 mmol/g, respectively. Therefore, doping metal in COFs is an effective method to improve CO_2 adsorption capacity. Wang et al. [57] reported a hybrid material combining MOFs and COFs. The Br-COFs shell was in situ grown on the surface of the NH_2-UiO-66 core to prepare a covalently connected core-shell NH_2-UiO-66 @Br-COFs hybrid material (Fig. 10). The material formed abundant ultramicropores at the core-shell interface, which created favorable conditions for CO_2 adsorption, and the unique structure of the core-shell interface can be adjusted by the coverage of Br-COFs.

2.2.5. Mesoporous M41s materials

Mesoporous materials are porous materials with uniform pore diameter between 2 and 50 nm according to the regulations of the International Union of Pure and Applied Chemistry (IUPAC) [58]. Mesoporous materials possess large specific surface area and pore volume, and the unique pore structure makes them have good adsorption capacity [59]. Compared with zeolites, mesoporous materials can adsorb macromolecules that are difficult to be adsorbed by zeolites. At the same time, mesoporous materials are also a kind of CO_2 adsorbent with good adsorption capacity. Among the mesoporous materials, M41s series has been the most studied and widely used. Since Kresge et al. [60] synthesized the mesoporous M41s materials for the first time in 1992, a lot of research on the M41s series have been carried out, and scholars have studied and discussed their CO_2 adsorption performance.

At present, many mesoporous M41s materials are modified by amine-impregnated to improve their CO_2 adsorption capacity. Mukherjee et al. [61] used monoethanolamine (MEA), benzylamine (BZA) and N-(2-aminoethyl) ethanolamine (AEEA) to modify MCM-41. The results showed that the adsorption effect of MCM-41 modified by AEEA was the best, and the CO_2 adsorption capacity of MCM-41–40% AEEA was 2.3 mmol/g, which was 3.5 times that before modification. In addition to pure amine impregnation, Son et al. [62] impregnated MCM-41 and MCM-48 with methanol solution containing 50 wt% polyethyleneimine (PEI), and found that the CO_2 adsorption capacity of these two materials was significantly higher than that of pure PEI impregnated sample. Both mesoporous materials showed reversible CO_2 adsorption-desorption behavior with recovery> 99%, and the adsorption effect of MCM-48 was better than that of MCM-41. Sharma et al. [63] added two mixed solutions of 30 wt% and 70 wt% PEI to impregnate MCM-41 and MCM-48. The experimental results are consistent with those of Son et al. It is further confirmed that MCM-48 is a mesoporous material more suitable for PEI impregnation modification than MCM-41. Based on this conclusion, Qian et al. [64] introduced cetyltrimethylammonium bromide (CTAB) additive. In the experiment, MCM-48 samples with long alkyl chain templates were successfully synthesized using CTAB as the guide template, and the MCM-48 with CTAB removed after calcination was labeled as MCM-48-C. The MCM-48 samples before and after calcination were impregnated with PEI to test their CO_2 adsorption performance (Fig. 11). The results indicated that

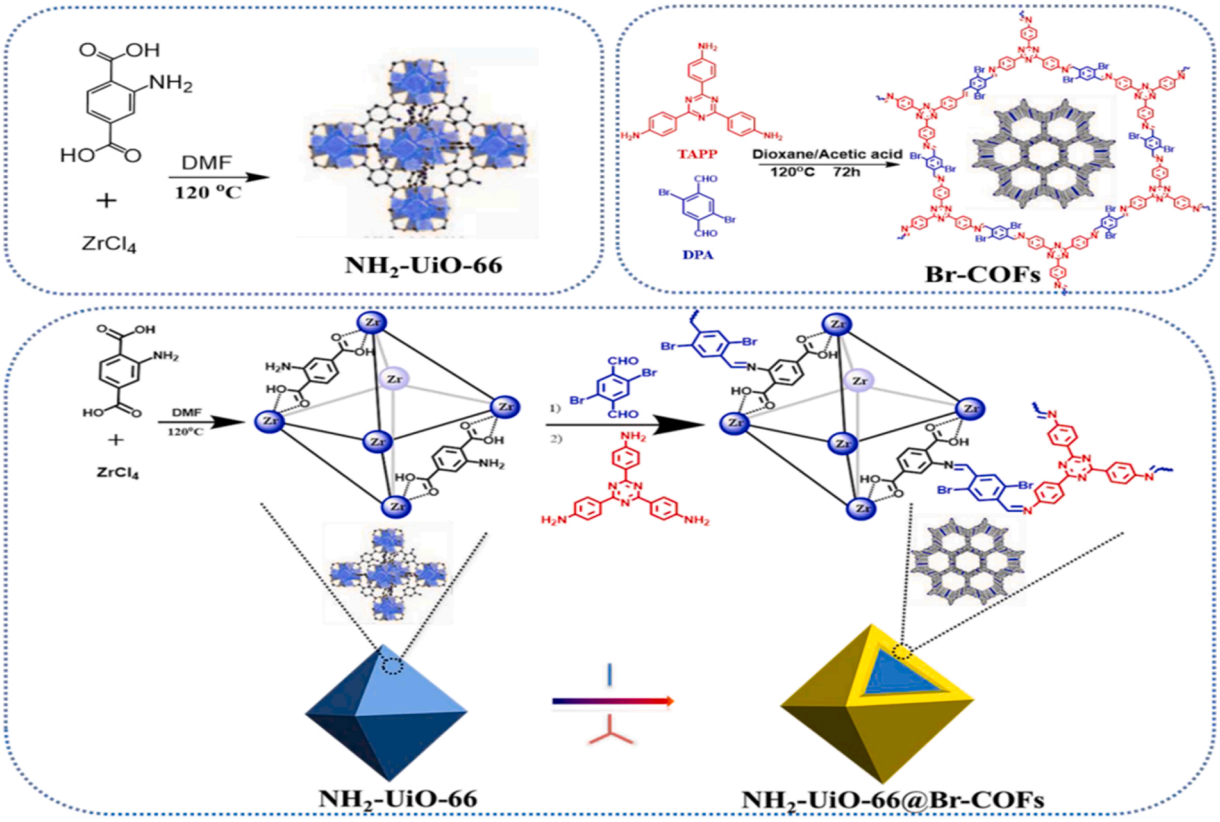


Fig. 10. Synthesis process of covalently connected core-shell NH_2-UiO-66 @COFs hybrid materials [57].

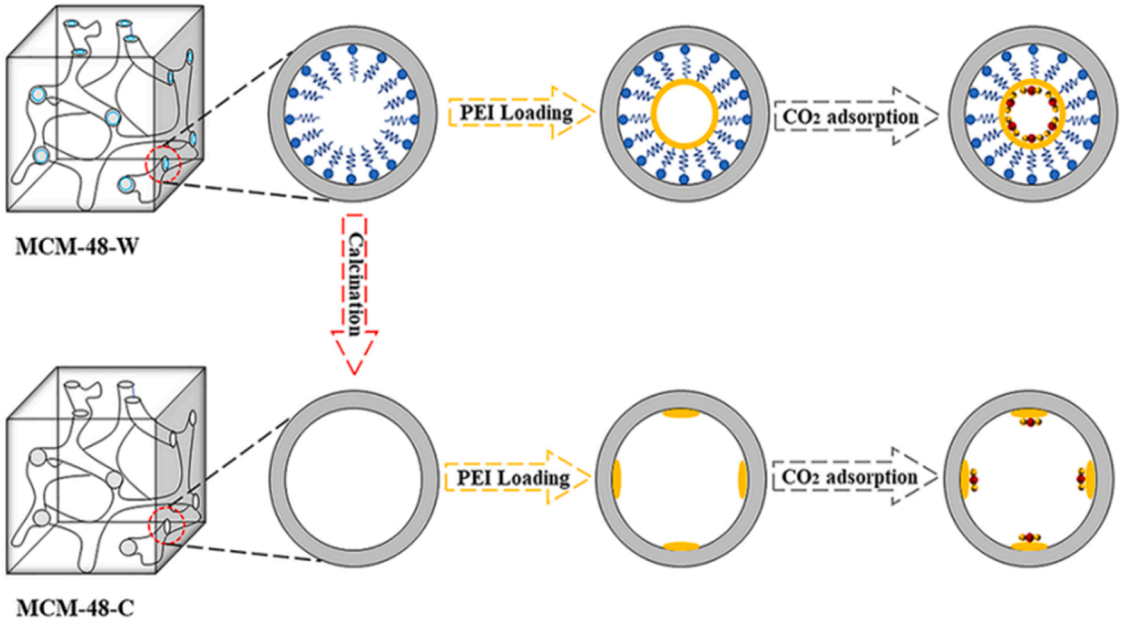


Fig. 11. Comparison of CO_2 adsorption by MCM-48 before and after calcination [64].

the addition of CTAB significantly improved the dispersibility of PEI and realized the efficient adsorption of CO_2 by the modified adsorbent. Compared with MCM-48-C, the pre-calcined MCM-48 (MCM-48-W) had a stronger adsorption capacity for CO_2, while expressing excellent regeneration ability.

In addition to amine impregnation, ionic liquids are also a good choice to improve the CO_2 capture ability of mesoporous materials. Huang et al. [65] fixed amino acid ionic liquid in mesoporous silica (PE-SBA-15), and the obtained supported adsorbent showed extremely good CO_2 capture ability under room temperature and pressure. This method may be combined with the mesoporous M41s materials, which is expected to provide a new strategy for CO_2 capture in air.

2.2.6. Hydrotalcites

Different from several adsorbents such as activated carbon and zeolite, hydrotalcites are high-temperature adsorbent [1]. In the temperature range of 200–400 °C, hydrotalcites show high CO_2 adsorption capacity and stability, and are easily regenerated by temperature or pressure swing [66,67]. The adsorption capacity of hydrotalcites at high temperature is related to the basicity of its surface. Therefore, alkali metal impregnation is a common method to improve the basicity of hydrotalcite and promote its adsorption of weakly acidic CO_2 [68].

Wu et al. [69] found that the potassium-promoted hydrotalcite-like adsorbent prepared with potassium nitrate (KNO_3) as the K precursor showed superior CO_2 adsorption capacity. Under the condition of 656 K, the CO_2 adsorption capacity of the hydrotalcite reaches 1.13 mmol/g. This result is larger than that reported in previous studies using potassium carbonate (K_2CO_3) as the K precursor. In addition to K, Faria et al. [68] found that the hydrotalcite (Mg^{2+}/Al^{3+} molar ratio of 2) modified by alkali metals Na and Cs also showed a high CO_2 adsorption capacity. Among them, the hydrotalcite doped with 9.4 wt% Cs showed the best adsorption performance.

2.2.7. Geopolymers

Geopolymers are novel zeolite-like inorganic cementitious material, usually prepared from aluminosilicate natural minerals or solid waste precursors, which exhibit properties similar to zeolites in terms of ion exchange or gas adsorption [70,71]. However, compared to zeolites, the internal structure and properties of geopolymers are related to the properties of synthetic raw materials, so that the design of geopolymers can be purposefully regulated [71,72]. The adsorption of CO_2 by geopolymers is an emerging technology recently, and the papers on the adsorption of gases, especially CO_2, by geopolymers are currently scarce.

Among many industrial wastes, coal ashes are good ingredients for synthetic geopolymers, and its chemical composition and production temperature are the key factors in determining the characteristics of geopolymers [72]. Freire et al. [31] were the first to use fly ash and rice husk ash to prepare geopolymers. It was found that the geopolymer prepared from calcined rice husk ash activated by NaOH exhibited excellent CO_2 adsorption performance, and its capacity was 24.4% higher than that of the best geopolymer adsorbent reported in the previous literature. In addition to the preparation of geopolymers, the synthesis of geopolymer composites has also attracted much attention. Chen et al. [73] prepared a phosphoric acid based geopolymer/activated carbon (PAG/AC) composite adsorbent exhibited higher CO_2 adsorption capacity than PAG at room temperature. Combined with the high-temperature adsorption properties of hydrotalcite, Papa et al. [74] synthesized a new geopolymer-hydrotalcite composite material suitable for CO_2 adsorption at intermediate temperature (200–400 °C). At 200 °C, the CO_2 adsorption capacity of the composites (0.109–0.145 mmol/g) was significantly higher than that of the geopolymer matrix (0.052 mmol/g). The CO_2 adsorption capacity of the composites mainly depends on the content of hydrotalcite.

2.2.8. CO_2 adsorption technology and its application in pilot plant

According to the regeneration method of the adsorbent, the adsorption process can be denominated as temperature swing adsorption (TSA), pressure swing adsorption (PSA) and electrical swing adsorption (ESA) [75–77]. TSA refers to the process in which the adsorbent adsorbs at room temperature or low temperature and desorbs at high temperature under the condition of constant pressure [78]. This process usually requires high desorption temperatures, which will increase the energy consumption during CO_2 capture. A VTSA technology that combines VSA (vacuum swing adsorption) with TSA effectively solves this problem and realizes the regeneration of the adsorbent at low temperature [79]. Compared with TSA, the regeneration conditions of VTSA are milder, which can greatly reduce the energy consumption and cost of CO_2 capture. PSA refers to the process of adsorption under pressure and desorption under reduced pressure or normal pressure under the condition of constant temperature [78]. When the desorption is in a vacuum state, it is also called vacuum pressure swing adsorption (VPSA). VPSA is a promising option for CO_2 capture, and its advantages such as low energy consumption, low cost, and high operational flexibility (depending on the applicable temperature range of the selected adsorbent) have attracted a lot of research efforts [80,81]. The essence of ESA is TSA, except that the desorption of ESA is promoted by applying electric current to generate heat and increasing the temperature through the Joule effect [82–84]. As one of the promising methods for CO_2 capture, ESA can significantly reduce the heating time and transfer the lost heat directly to the adsorbent material compared to TSA and ESA, thereby increasing process productivity and minimizing heat loss [83,85]. The main disadvantage of ESA technology is the consumption of electrical energy in the process of capturing CO_2. Therefore, the feasibility of implementing this technology depends on whether the value generated per unit CO_2 captured can compensate for the energy consumption in the capture process [85].

So far, there are not many reports on the application of the above-mentioned adsorption technologies for CO_2 capture in pilot plants. Therefore, it is necessary to conduct pilot-scale experiments to evaluate practical CO_2 separation performance and capture energy consumption. Plaza et al. [86] proposed a post-combustion CO_2 capture process using carbon honeycomb monoliths and steam stripping TSA, and took an 800 MWe supercritical coal-fired power plant as a reference case to evaluate the performance of TSA technology applied to post-combustion CO_2 capture process. Cheng et al. [87] utilized a three-bed, nine-step PSA process to capture CO_2 from flue gas obtained after desulfurization and water removal from a coal-fired power plant, and used a central composite design (CCD) to determine the optimal operating conditions in the process. The simulation results showed that the CO_2 purity at the bottom of the adsorption tower could reach 89.20% and the recovery rate was 88.20%. Liu et al. [88] studied a three-bed seven-step VPSA process for CO_2 capture from flue gas emitted from coal-fired power plants, which included flushing and pressure equalization steps. At a feed flow rate of 46.0 Nm^3/h, 85% CO_2 was captured with recovery of 79% from the flue gas. Wang et al. [89] have done similar work. Using a two-stage VPSA unit, they experimentally evaluated the capture of CO_2 from the flue gas of an existing coal-fired power plant, and at a feed flow rate of 35 Nm^3/h, 90.2% CO_2 could be recovered with a purity of 95.6%. Grande et al. [90] assessed the technical feasibility of ESA technology to capture CO_2 from flue gases of natural gas power stations where the CO_2 content is 3.5%. The sorbent consisted of a honeycomb monolith with 70% zeolite 13X and 30% binder conductive material to ensure a continuous medium for electricity across the adsorbent. The simulation results showed that the recovery of CO_2 was 72% and the purity was 89.7%. This work demonstrates that ESA can be one of the CO_2 capture technologies worthy of further exploitation.

2.3. Membrane separation

The gas membrane separation technology is a new generation of gas separation technology developed in the 1980 s. The principle of action is to realize gas separation under pressure driven by the difference in the permeation rate of various gas components on the surface of the polymer membrane, as shown in Fig. 12 [91]. Membranes can be classified into two categories: polymer membranes (organic membranes) and inorganic membranes. At present, polymer membranes have been involved in many industries for CO_2 separation. But their selectivity for CO_2 is not ideal, which is not suitable for large-scale CO_2 separation and purification [1,92]. In contrast, inorganic membranes exhibit better stability, corrosion resistance, and higher separation performance, making them a promising separation membrane material [92]. However, inorganic membranes are rarely used in industry due to the high cost.

In recent years, the membrane separation method, as a high technology, has attracted much attention in the CO_2 separation process.

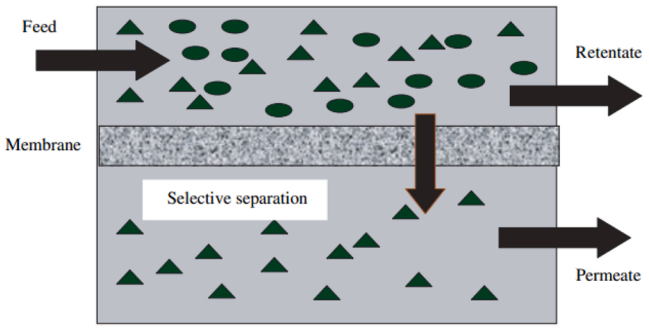


Fig. 12. Schematic diagram of gas separation membrane [93].

Compared with other traditional CO_2 separation technologies, membrane separation has the characteristics of low energy consumption, low cost, small footprint, no pollution, and simple process. It is a promising CO_2 capture technology [94]. However, the separation efficiency of the membrane separation method is low, and it is difficult to obtain high-purity CO_2. Therefore, in the future research on membrane separation technology, the key to promote the CO_2 separation efficiency is improving the membrane materials [95].

Prasad et al. [96] added polyallylamine (PAA) to chitosan (CS) to make CS / PAA blend membrane to capture CO_2 and achieved good CO_2 separation effect. Borgohain et al. [97] found that the blend membrane prepared by carboxymethylchitosan (CMC), carboxymethylated derivative of chitosan, after adding non-volatile polyamidoamine (PAMAM) dendrimers also expressed a good CO_2 separation effect. When the temperature was 90 ℃, the blend membrane containing 10 wt% PAMAM had the highest CO_2 permeability and CO_2/N_2 selectivity.

As one of the inorganic membranes with good performance, ceramic membrane shows excellent CO_2 separation ability. At present, more research is to use ceramic carbonate biphasic membrane to separate CO_2 at high temperatures. Ovalle-Encinia et al. [98] prepared a ceramic-carbonate film by using $Ce_{0.85}Sm_{0.15}O_2$-$Sm_{0.6}Sr_{0.4}Al_{0.3}Fe_{0.7}O_3$((SDC-SSAF) composite material. This membrane material exhibited eximious CO_2 separation ability at high temperatures (700–900 °C). Chen et al. [99] synthesized a $Ce_{0.8}Sm_{0.2}O_{2-\delta}$(SDC)-carbonate biphasic membrane that showed excellent CO_2 permeation flux, up to 5.46 ml min^{-1} cm^{-2}. Under 50%CO_2 − 50%N_2 feeding condition, the stabilization time of the membrane was 85 h. Zhang et al. [100] explored a new ceramic-carbonate biphasic MOCC membrane composed of $BaZr_{0.8}Y_{0.2}O_3$-δ (BZY) and Li_2CO_3–Na_2CO_3 mixed crystals with excellent chemical stability. After adding 3% H_2O to the purge gas, the stabilization time of the membrane could reach 250 h. At high temperatures (550–750 ℃), the maximum CO_2 permeation flux of the membrane material was 0.53 ml min^{-1} cm^{-2}.

2.4. Cryogenic separation

Cryogenic separation is a process in which solid CO_2 is separated from the gas mixture by desublimation under low-temperature conditions (temperature is less than −56.6 ℃), which is a physical separation process [101,102]. The most prominent feature of this process is that it can melt solid CO_2 and recover CO_2 in the form of high-purity liquid, which is convenient for injection into underground storage or utilization (such as injection into formations to enhance oil recovery) [93,103]. Although the purity of CO_2 obtained by the cryogenic separation method is very high, it is generally only applicable to the case of high CO_2 concentration [93]. For flue gas with high temperature, low CO_2 concentration and low partial pressure, the cryogenic separation method is not suitable [92]. The traditional low-temperature separation technology consumes a lot of energy in the process of compression and condensation, and when the temperature is lowered to a certain level, the CO_2 will solidify, causing the pipeline to be blocked and affecting the separation of the gas [104]. Overall, the separation process is relatively complex, the equipment investment is high, and it has not been put into use on a large scale.

Yousef et al. [105] optimized and improved the traditional low-temperature CO_2/CH_4 separation model, overcame the pipeline blockage caused by CO_2 freezing and precipitation, and extracted liquid CO_2 with purity up to 99.92% from biogas (mainly a mixture of CO_2 and CH_4). The model has a good application prospect in the separation of CO_2 from biogas. Babar et al. [106] proposed a new cryogenic packed bed system. The system could continuously capture CO_2 in natural gas at low temperature, which improved the problems of dry ice formation and blockage of cryogenic packed bed caused by CO_2 condensation. Sateesh et al. [107] designed a solar integrated vapor absorption refrigeration (VAR) system, which effectively reduced the energy consumption of CO_2 separation. The designed solar VAR system provided the energy required for the CO_2 condensation process. Under certain conditions, the energy consumption of CO_2 separation could be reduced by up to 37.54%.

2.5. Hydrate method

Gas hydrate is a cage shaped crystal envelope formed under the action of hydrogen bond of water, which is composed of selective combination of host molecule (H_2O) and guest molecule [108,109] (Fig. 13). Hydrate method is a new gas separation technology. Its separation principle is the great difference in the pressure of gas hydrate formed by various gases and hydrates. The components with low pressure are enriched in the hydrate phase, and the components with relatively high pressure stay in the reactor, so as to realize the separation of mixed gases [110,111]. At 0 °C, the formation pressure of CO_2 hydrate is about 0.52 MPa. When CO_2 is separated by the hydrate method, the hydrate envelops CO_2 in cage-shaped pores, and one pore can only accommodate one CO_2 molecule. Under standard temperature and pressure, 1 m^3 of hydrate can accommodate 160–180 m^3 of CO_2 [110]. The separation of CO_2 by the hydrate method has the characteristics of no pollution, no corrosion, simple process and low energy consumption. Among the new CO_2 capture technologies, the hydrate method expresses strong competitiveness [112,113].

Wang et al. [113] used tetra-n-butyl ammonium bromide (TBAB) semi-clathrate hydrate to capture CO_2 in unconventional natural gas, as shown in Fig. 14, and investigated the CO_2 capture effect of TBAB at different concentrations. The results showed that 2.57 mol% TBAB had

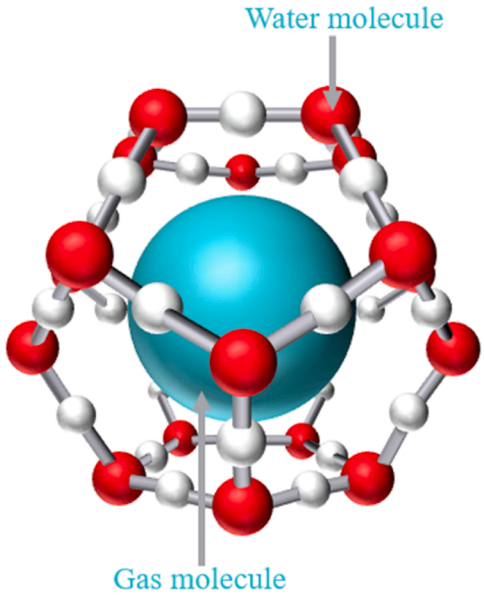


Fig. 13. Schematic diagram of gas hydrate.

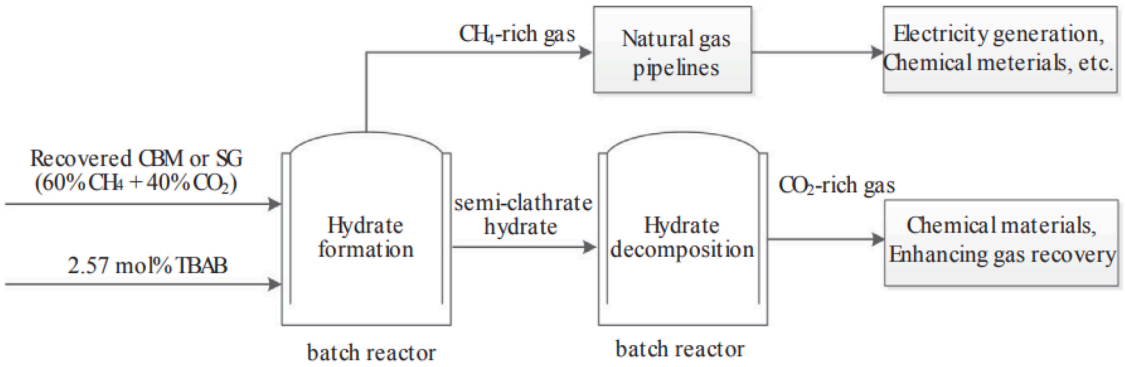


Fig. 14. Flow chart of CO_2 capture from unconventional natural gas using TBAB semi cage hydrate [113].

the best effect on CO_2 capture. The addition of sodium dodecyl sulfate (SDS) promoted the formation of hydrate, but reduced the separation efficiency of CO_2. Cheng et al. [114] added SiO_2/Al_2O_3 nanoparticles (NPs) to TBAB, which accelerated the formation of hydrate and enhanced the CO_2 capture ability of hydrate. Compared with the SDS-TBAB reaction system, the mixture of NPs and TBAB exhibited higher CO_2 capture capacity. Therefore, nanoparticles have certain potential in replacing traditional additives and promoting the capture of CO_2 by hydrate. Costa et al. [115] reported a method for the continuous production of CO_2 hydrates using NetMIX, a novel mixing technology. Importantly, this technology overcomes the technical limitations caused by mass and heat transfer in large-scale CO_2 capture, which is crucial for the industrial development of CO_2 capture.

2.6. Microbiological method

Microorganisms are a collective term for all tiny organisms that are difficult to be observed by the naked eye. The fundamental principle of microbial capture of CO_2 is to use the photosynthesis of microorganisms to absorb and fix CO_2. It has the characteristics of low energy consumption, no corrosion, green environmental protection, etc., and can directly convert CO_2 into organic matter, but it needs large-scale cultivation to achieve good results [116,117]. According to statistics, the net carbon sequestration of autotrophic microorganisms can reach 7 billion tons per year, which indicates that microbial carbon sequestration is indeed an effective method to improve global climate change [118]. As one of the emerging CO_2 capture technologies, it fully conforms to the concept of green environmental protection, non-toxic and harmless, and shows a good development prospect.

Microalgae have become a good choice for microbial carbon sequestration due to their strong reproductive capacity, fast photosynthetic rate, and short growth cycle. However, the carbon sequestration rate of traditional microalgae fixation method is only 5–20% [119]. Song et al. [120] proposed a novel absorption-type microalgae composite CO_2 capture process, which combined chemical absorption with microbial carbon sequestration, greatly improving the carbon sequestration rate. In this process, ammonia-rich solution was used to absorb CO_2 to generate NH_4HCO_3, and then the liquid containing NH_4HCO_3 was put into the microalgae pond for biotransformation. Microalgae culture system was used to replace the traditional analytical column, which not only avoided high energy loss, but also obtained organic products with high added value. Three microalgae, L166, L38 and UTEX1602, with high potential for chemical-biological trapping were screened out. Its carbon sequestration rate could reach above 85%.

2.7. Application of CO2 capture technologies in basic industries

A large amount of CO_2 emissions from basic industries mainly comes from the consumption of fossil fuels, most of which comes from the iron, steel, cement and chemical industries [121,122]. As one of the main industrial sources of CO_2 emissions, the steel, iron and cement industries, CO_2 emissions reduction in the production process has always been a research hotspot in academia and industry.

In 2020, the iron and steel industry have emitted about 2.61 billion tons of CO_2, an estimated 7.5% of total global emissions [123]. At the same time, CO_2 emissions from the cement industry reached 2.53 billion tons in 2020, accounting for an estimated 7.27% of global emissions, up two percent from 2018 [121,123]. The flue gas produced by the iron, steel and cement industries has a similar CO_2 content of about 20–30%, and the applicable capture technologies are roughly the same [124]. At present, the research on CO_2 capture in the iron, steel and cement industries mainly focuses on several post-combustion capture technologies.

Judging from the existing post-combustion capture technologies, chemical absorption is undoubtedly the best choice. Dreillard et al. [125] explored the application of amine scrubbing technology to capture CO_2 in the iron and steel industry in three processes: Hicapt™, Hicapt+ ™ and DMX™. Li et al. [126] developed three innovative configurations of the MEA-based CO_2 capture process that can reduce the regeneration energy of the absorbent by up to 22%. Although most adsorbents are less selective for CO_2, this does not affect the fact that solid adsorption is a good choice as flue gas from iron, steel and cement industries has a relatively high CO_2 concentration [123]. In 2020, Svante, LafargeHolcim, Oxy Low Carbon Ventures and Total launched a joint research project to assess the commercial viability of an adsorption technology-based carbon capture facility at LafargeHolcim's Portland Cement Plant in the United States [127]. With an annual CO_2 capture capacity of about 20 tons, the project is expected to become the largest carbon capture plant in the cement industry. Membrane separation is theoretically feasible for the capture of flue gas from iron, steel and cement plants, which mainly depends on the partial pressure of each component in the flue gas and the selectivity of the membrane [127]. In a recent study, Janakiram et al. [128] conducted CO_2 capture tests on flue gas from the Colacem cement plant in Italy using a hybrid facilitated transport membrane. Wu et al. [124] designed a multifunctional two-stage membrane device to conduct research on CO_2 capture in steel mills and cement plants. Calcium looping is one of the most promising CO_2 capture technologies in the cement industry, which utilizes lime (CaO) to react with CO_2 in flue gas at high temperature to form calcium carbonate. This technology is currently being used in pilot-scale coal--fired fluidized bed combustors and cement production [121].

In addition to the iron, steel and cement industries, the chemical, pulp, paper, glass and ceramic industries are also important research targets for carbon reduction strategies. The CO_2 capture technology concentrated in the chemical industry is mainly chemical absorption, most of which are based on amine solvents [129]. The pulp and paper industry has huge carbon capture and storage potential, with most of its CO_2 emissions from on-site biomass burning [130]. Hektor et al. [131] evaluated the feasibility of CO_2 capture in kraft pulp and paper mills using MEA and chilled ammonia (CA). Similar to Hektor et al., Nwaoha

et al. [132] evaluated the economics of chemical absorption for CO_2 capture from a pulp mill using two advanced configurations. Regarding the glass and ceramic industry, there are few reports on the application of CO_2 capture technology. There are probably two main reasons for this. First, from a technical point of view, most ceramic processes have less waste-gas streams than those of other industries such as cement and steel, and the CO_2 content in the waste-gas streams is too low for effective carbon capture [133]. Second, from an economic point of view, glass and ceramic manufacturing is often located in remote and rural areas, which will impose extremely high costs on the development, operation and subsequent transportation of carbon capture emissions systems [134]. Therefore, CO_2 capture technology may not be suitable for application in the glass and ceramic industry.

2.8. Critical comparison of CO2 capture technologies

Among the above CO_2 capture technologies, each has its advantages and certain limitations (Table 4). This subsection compares and analyzes several capture technologies, and proposes future prospects and suggested directions for different technologies.

In this series of capture technologies, the one with the highest CO_2 capture capacity and the most suitable for industrial applications is chemical absorption. Although solid adsorption has a lower capture capacity than chemical absorption, the technology has good potential for industrial deployment based on the application of several adsorption technologies in pilot plants. The CO_2 obtained by low temperature separation is very pure, but its process is too complicated and its cost is high, so it is not used much in industry. Membrane separation, hydrate method and Microbiological method all have certain advantages in economy and environmental protection, but there is still a long way to go to realize industrial application. More research is needed to solve existing problems and realize industrial deployment as soon as possible.

3. CO_2 utilization

After CO_2 is captured, how to deal with CO_2 is also a question that many people concerned. A large number of CCS projects have been launched around the world to store captured CO_2. However, due to the high cost and uncertainty of carbon storage (such as the potential leakage of CO_2 stored in the ground and the possibility of small earthquakes caused by CO_2 under pressure), there is an increasing preference for CCUS technology that can reuse CO_2 [135]. Therefore, the "U" in CCUS has become an important direction to deal with CO_2. Compared with CO_2 storage, the vast majority of CO_2 utilization cannot permanently remove CO_2 from the atmosphere to achieve negative emissions [136]. Furthermore, it should be clear that net-zero-CO_2 emissions are unlikely to be achieved with CO_2 utilization alone, as the electricity consumed in the process of CO_2 utilization will increase CO_2 emissions [136]. However, from the perspective of long-term stability and economic benefits, the utilization of CO_2 also plays a pivotal role in addressing the CO_2 problem. Specifically, "U" is defined as the resource utilization of CO_2, which mainly includes physical utilization, chemical utilization, biological utilization, and mineralization utilization [1,3].

3.1. Physical utilization

Physical utilization of CO_2 is very widely used, mainly in food, refrigeration, petroleum and other industries. What we need to know is that almost all physical uses (except CO_2 flooding) just delay the release of CO_2, which is eventually released into the atmosphere [137]. It follows that most physical utilization does not sequester CO_2 from the environment and is not a decarbonization technology. But the economic benefits of physical utilization will lay the foundation for future carbon capture and storage projects. This article provides an overview of several common physics applications.

Table 4
Comparison and analysis of CO_2 capture technologies [29,92,93,95,112,116,120].

Capture technology	Advantages	Limitations	Future outlooks
Chemical absorption	Superior CO_2 capture performance; excellent CO_2 selectivity; relatively mature technology	High corrosion rate of equipment; high energy consumption and high operating costs; discharge of absorbed solvents will cause environmental pollution	Develop new absorbent solvents with good absorption performance and low corrosion; Develop equipment and absorption processes with low desorption energy consumption
Solid adsorption	Low energy consumption for regeneration; almost non-corrosive; suitable for low-concentration CO_2 capture; relatively simple operation; multiple adsorption technologies (TSA, PSA, ESA)	The capture capacity is smaller than chemical absorption; the capture of industrial flue gas needs to improve the cycle stability	In order to further improve the CO_2 capture performance, it is necessary to modify the adsorption material; optimization and combination of several adsorption technologies
Membrane separation	Low cost; low energy consumption; simple operation process; high separation purity ; be friendly to environment	Most membranes are easily damaged and has a short service life; the separation purity is not high; moisture has a greater impact on the permeability of the polymer membrane; industrial applications require a larger membrane area	Develop advanced new membrane materials to improve CO_2 separation efficiency; more industrial pilot research needs to be invested
Cryogenic separation	It is suitable for the capture of CO_2 with high concentration and high partial pressure, and the separation purity is very high	High operating cost and complex process; the solidification of CO_2 at low temperature leads to pipeline blockage; not suitable for the capture of low CO_2 volume fraction	It is necessary to design and optimize a more effective, simple low-temperature process, and solve the problem of CO_2 solidification blocking the pipeline
Hydrate method	Low energy consumption; non-corrosive and non-polluting; simple process flow	Few practical applications and immature technology; high requirements for equipment selection; CO_2 separation efficiency is affected by the hydrate formation method in a specific pressure vessel	More in-depth research is needed to improve CO_2 separation efficiency and develop new equipment and processes
Microbiological method	No pollution; low cost; high return value; no complicated process required, can directly convert CO_2 into organic matter	There are few related researches and it is still in the theoretical stage	More experimental studies are needed to promote the development of microbiological methods for CO_2 capture

3.1.1. Beer and carbonated beverages

Beer and carbonate beverages have been major applications of CO_2 for a long time. According to statistics, in 2021, the production of beer and carbonated beverages in China are about 35.6 million tons and 23.6 million tons, respectively. Based on the average consumption of 19 kg CO_2 per ton of beer and 17 kg CO_2 per ton of carbonated beverages, the average annual demand for CO_2 are about 1078,000 tons. Under certain conditions, CO_2 injected into beer and beverages can bring different tastes, and CO_2 can also play a role in anti-corrosion. Compared with other types of beverages, carbonated beverages and sparkling wine beverages have a longer expiration date [138]. At present, with the emergence of different beverages, the growth rate of carbonated beverages and beer industries has been significantly reduced, and the use of CO_2 in the industry has gradually decreased.

3.1.2. Fresh frozen of food

At present, CO_2 has been widely used in the fresh frozen of food. By adjusting the concentration ratio of oxygen and CO_2 in the storage environment, it is beneficial to the freshness of fruits and vegetables [138]. Injecting a certain amount of CO_2 can inhibit the reproduction of bacteria and mitigate the rot of food, thereby better maintain the freshness of fruits and vegetables [138]. Solid CO_2 (dry ice) works well for cold storage. The sublimation of dry ice produces low-temperature CO_2 gas, which can isolate the food from air, so as to achieve the effect of lasting refrigeration [138]. Due to the high cost of cold storage, dry ice is often used for cold storage and transportation of high-grade food. Parton et al. [139] designed a continuous device for inactivating microorganisms by high-pressure CO_2, which can be used for fresh food storage. In the experiment, microbial strains were used as test microorganisms, and liquid food was used to test the sterilization effect of the device. The results showed that the device exhibited good sterilization effect on both natural red orange juice and commercial tomato sauce. Bacillus subtilis could be completely inactivated by inoculation in physiological saline. Ahn et al. [140] used CO_2 to treat 'Seolhyang' strawberries to explore whether CO_2 affected the quality of strawberries while prolonging their expiration date of strawberries. The experimental results indicated that the hardness, color and metabolite characteristics of CO_2-treated strawberries were almost the same as those of untreated strawberries. CO_2 did not reduce the quality of strawberries while prolonging the expiration date of strawberries. Therefore, CO_2 has certain potential advantages in the cold storage of food.

3.1.3. CO_2 gas shielded welding

CO_2 gas shielded welding is a welding method in which CO_2 is used as shielding gas in the outer layer. Its operation is simple, but it is limited to working in the absence of wind [141]. Compared with the traditional manual arc welding method, CO_2 gas shielded welding has the advantages of high working efficiency, low electricity consumption, energy saving, low cost and easy automation. With the advancement of science and technology, CO_2 gas shielded welding technology has also been significantly improved, but there are still some related problems. For example, due to the quality of welding materials, it is easy to cause splashes and pores during the working process, resulting in poor welding effect [142,143]. M. Wahba et al. [144] used 100% CO_2 shielding gas to reduce the cost of fiber laser-GMA hybrid welding of double-side welded T-joints, limiting the generation of pores. But it resulted in increased spatter. Pan et al. [145] used 80%Ar–20%CO_2 and 100% CO_2 shielding gas to weld thick plates of high-tensile-strength steel, HT780. The results were consistent with the conclusion of M. Wahba et al., that is, using 100% CO_2 shielding gas would produce more spatters. But pan et al. found that the generation of spatter could be reduced by adjusting the arc parameters of 100% CO_2 shielding welding.

3.1.4. CO_2 flooding

CO_2 flooding is the process of injecting CO_2 into the formation to increase oil fluidity and enhance oil recovery. In the early days of the oil industry, people only relied on the natural energy of the formation to extract oil (primary oil recovery), but the extracted oil was less than 20% of the underground reserves [146]. With the development and progress of the oil industry, it has been found that some oil can be extracted (secondary oil recovery) by injecting water into the formation, so that the crude oil recovery rate can be increased to 20%− 40%, but there is still most of the oil in the formation can not be excavated [147]. Until 1952, several engineers in the United States proposed CO_2 flooding technology (tertiary oil recovery). This technology injects CO_2 into the formation. Under the condition of high temperature and high pressure, the mixing of CO_2 and oil reduces the viscosity of the oil and increases the fluidity of the oil. CO_2 can penetrate into some tiny pores, which can better displace oil [148]. This extraction technique can increase oil recovery up to 60%− 70% [148]. The CO_2 flooding process is shown in Fig. 15.

CO_2 flooding can be divided into miscible flooding and immiscible flooding. CO_2 miscible flooding refers to the process of CO_2 extraction and crude oil gasification, which makes the interfacial tension between CO_2 and crude oil tend to zero, and then a miscible phase is formed [150]. During this process, the system pressure is much higher than the minimum miscibility pressure (MMP) [150]. The theoretical oil displacement efficiency of this oil displacement method can reach 100% [151]. The CO_2 miscible flooding process is represented in Fig. 16a. CO_2 immiscible flooding refers to the process in which the oil-gas interface always exists in the process of oil displacement, and the CO_2 injected into the reservoir drives the crude oil to migrate to the production wells on both sides. The system pressure of this process is generally lower than that of MMP, and the oil recovery efficiency is generally lower than that of miscible flooding [150]. The CO_2 immiscible flooding process is shown in Fig. 16b. Compared with other flooding methods, CO_2 flooding has the advantages of wide application range, low cost, and significantly improved oil recovery [152]. However, there are also some problems with CO_2 flooding. CO_2 has high fluidity and is not easy to control, and fingering phenomenon is prone to occur in the process of oil displacement, which reduces the storage efficiency of CO_2 [147]. When crude oil mixed with CO_2 is extracted from the reservoir to the ground and degassed, how to effectively prevent CO_2 from returning to the atmosphere again is also a issue we need to consider [147]. Chen et al. [150] studied the influence of gravity separation on CO_2 sequestration and crude oil production during CO_2 flooding under miscible and immiscible conditions, respectively. The experimental results revealed that the crude oil recovery of CO_2 flooding was seriously affected by gravity under immiscible conditions, but not obviously under miscible conditions. Gravity separation was conducive to weakening the fingering phenomenon in the oil displacement process and increased the CO_2 storage potential, but the influence of gravity on CO_2 storage decreased with the increase of pressure.

3.1.5. Tobacco expansion technology

CO_2 tobacco expansion technology is the process of expanding the shredded tobacco by injecting liquid CO_2 into the tobacco cells under a certain pressure, and then depressurizing to release the shredded tobacco [155]. This technology not only improves the physical and sensory quality of tobacco, but also saves an average of 5%− 6% of cut tobacco per box of cigarettes, reducing the cost of cigarette manufacturing [156]. China's total cigarette output in 2021 was about 48 million boxes. If all cigarette factories use CO_2 as expander, the total amount of CO_2 used by the tobacco industry will increase significantly each year. Therefore, CO_2 has a good application prospect in the tobacco industry.

Low boiling point and volatility of CO_2 can effectively reduce the harmful components and tar content in cigarettes, thus producing high value-added low tar cigarettes [155]. However, after expansion, some aromas in tobacco are taken away by CO_2, which reduces the aroma and taste of tobacco [155]. Therefore, it is necessary to improve the expansion process, such as adding flavors, bio-alcoholization, etc, which

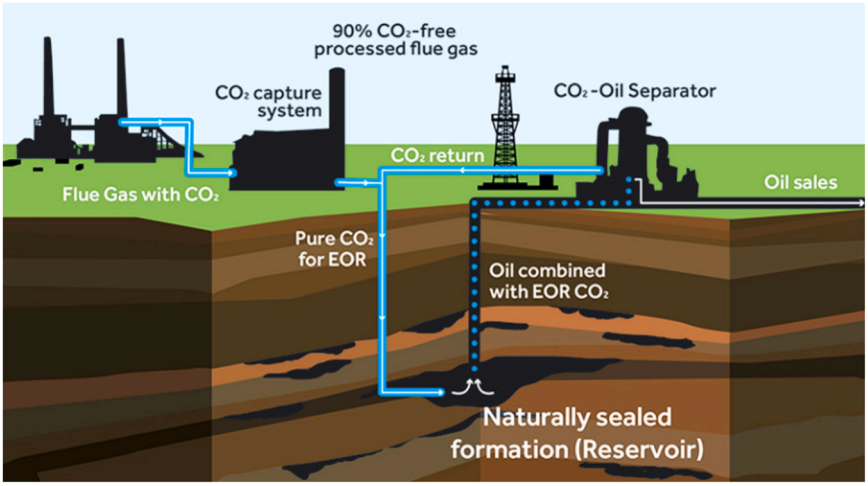


Fig. 15. Schematic diagram of CO_2 enhanced oil recovery process [149].

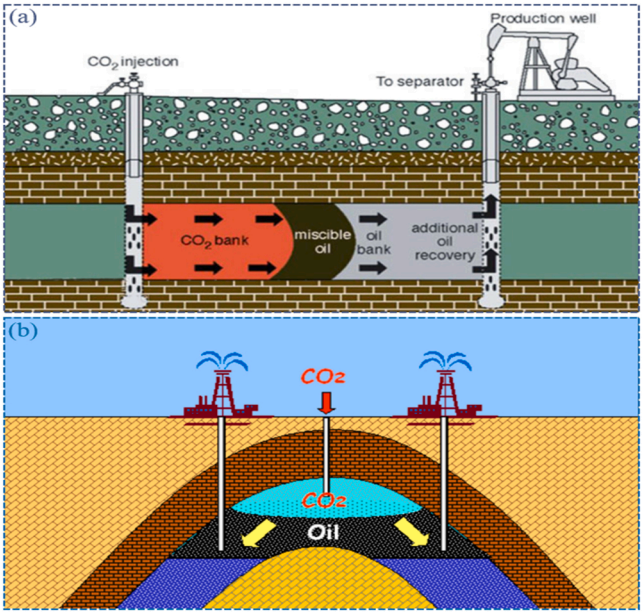


Fig. 16. (a) Schematic diagram of CO_2 miscible oil displacement (b) Schematic diagram of CO_2 immiscible oil displacement [153,154].

is usually an important technical means to further improve the sensory quality and applicability of expanded tobacco [157]. Recently, some studies have pointed out that the addition of enzymes can increase the aroma and improve the sensory quality of tobacco leaves [157–159]. This finding may be a new technical means to improve the quality of tobacco if combined with CO_2 expansion technology.

3.1.6. *Supercritical CO_2 cleaning agent*

Supercritical CO_2 is a cleaning agent with very good properties, which is non-toxic, harmless, odorless and non-corrosive [160]. Its cleaning principle is to use the dissolution characteristics of supercritical CO_2. Supercritical CO_2 fluid is a weakly polar solvent, which can dissolve non-polar organic compounds to a great extent and effectively remove weakly polar organic pollutants on the surface of items. For insoluble polar molecules, the cleaning effect can also be improved by adding cosolvent [161]. Gao et al. [162] adopted supercritical CO_2 to clean silicon wafers and performed supercritical drying to replace the previous semiconductor IC cleaning technology. Supercritical CO_2 cleaning avoided the problems of large water consumption, difficult drying and secondary pollution brought by semiconductor IC cleaning technology, and there was no adhesion phenomenon after supercritical drying. Lu et al. [163] combined ultrasonic cleaning technology with supercritical CO_2 cleaning technology to design an ultrasonic enhanced supercritical CO_2 composite cleaning experimental platform (Fig. 17), which greatly improved the cleaning efficiency. The experimental results revealed that the rust and oil stains on the surface of carbon steel plate could be removed under appropriate ultrasonic intensity, and both had an optimal cleaning time. Yang et al. [164] used supercritical CO_2 to clean uranium, and investigated some common residual liquids (water, engine oil and triethanolamine). The results indicated that supercritical CO_2 showed a good cleaning effect on water and engine oil, but it was necessary to combine ultrasonic cleaning with triethanolamine cleaning. Moreover, the corrosion resistance of uranium soaked in supercritical CO_2 was greatly improved.

3.1.7. *Supercritical CO_2 extractant*

Supercritical fluid extraction is a separation technology that emerged in the 1970 s, among which supercritical CO_2 extraction is the most widely used [165]. Supercritical CO_2 extraction has the characteristics of non-polluting, safety, non-toxicity, no combustion, and low price, and has significant advantages over traditional extraction techniques [165]. At present, supercritical CO_2 extraction technology has been involved in medicine, food, cosmetics, biological industries, etc. [166,167].

Devani et al. [168] used the central composite rotatable design (CCRD) to optimize the design of supercritical CO_2 extraction, and successfully shortened the extraction time of onion oleoresin. Coelho et al. [169] added co-solvents isopropanol, ethanol, and ethyl lactate to extract spent coffee grounds (SCGs). Compared with pure supercritical CO_2, the addition of co-solvents reduced the time required to extract maximum oil yield by half. Leila et al. [170] used supercritical CO_2 to extract oil from Algerian Inula Viscosa leaves, and found that the increase of pressure could significantly improve the extraction yield of oil.

3.1.8. *Supercritical CO_2 fracturing*

Supercritical CO_2 can be used as a new fracturing medium for fracturing unconventional oil and gas reservoirs [171]. This is a brand-new anhydrous fracturing process and a special form of CO_2 dry fracturing technology [172,173]. Compared with other fracturing processes (hydraulic fracturing, CO_2 dry fracturing), supercritical CO_2 fracturing technology has many advantages. The supercritical fluid is water free, which will not cause the expansion of the reservoir clay, nor will it damage the water-sensitive formation; Supercritical CO_2 has excellent solvation ability, which can dissolve the heavy oil components near the well; After fracturing, supercritical CO_2 will be rapidly reversed, saving

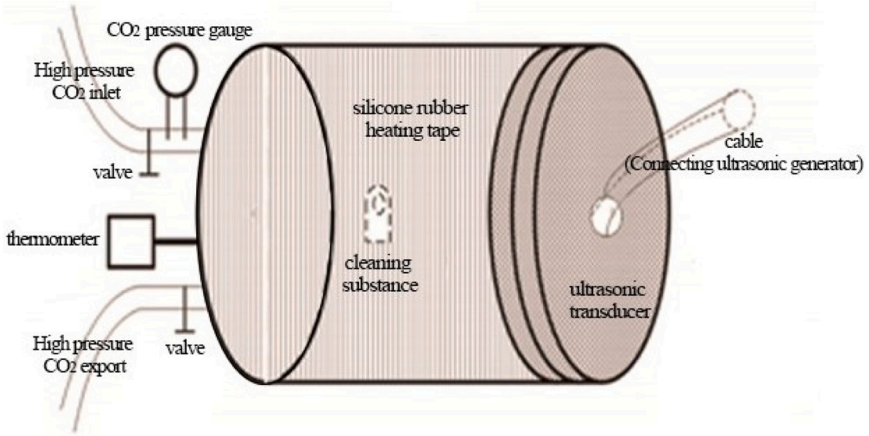
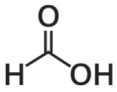
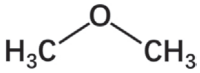
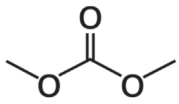
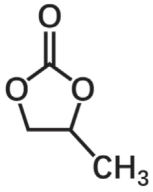
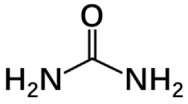
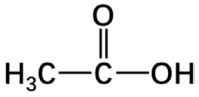
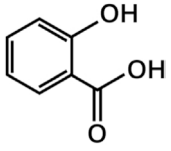
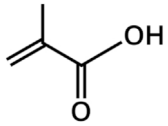


Fig. 17. Schematic diagram of the composite cleaning experimental platform [163].

the non-production time of oil and gas wells; Supercritical CO_2 can effectively replace methane adsorbed in shale, improve oil recovery and achieve the purpose of permanent CO_2 storage [172–174]. Therefore, supercritical CO_2 fracturing is considered as a promising anhydrous fracturing technology.

At the same time, the supercritical CO_2 fracturing technology also has some defects. The viscosity of supercritical CO_2 (0.02–0.26 mPa·s) is much lower than that of traditional fracturing fluid, and its ability to carry proppant as fracturing fluid has been controversial [172]. The lower viscosity makes the proppant prone to sedimentation, and it is

Table 5
Some chemicals that can be produced from CO_2 [1,175,177–179,182].

Chemical	Molecular formula	Synthesis method	Reaction classification
Methane	CH_4	Catalytic synthesis of CO_2 and H_2	Catalytic hydrogenation
Methanol	CH_3OH	Catalytic synthesis of CO_2 and H_2	Catalytic hydrogenation
Formic acid		Catalytic synthesis of CO_2 and H_2	Catalytic hydrogenation
Dimethyl ether		Direct synthesis of CO_2 and H_2 or dehydration of methanol generated from CO_2 and H_2	Catalytic hydrogenation
Dimethyl Carbonate		Catalytic Synthesis of CO_2 and Methanol	Esterification
Propylene carbonate		Catalytic Synthesis of CO_2 and Propylene Oxide	Esterification
Polycarbonate	–	Copolymerization of CO_2 with epoxide	Copolymerization
Polyurethane	–	Copolymerization of aziridine with CO_2	Copolymerization
Urea		Catalytic synthesis of CO_2 and NH_3 at high temperature and high pressure	Ammoniation
Acetic acid		Catalytic synthesis of CO_2 and methane	Carboxylation
Salicylic acid		Catalytic synthesis of CO_2 and phenol at high temperature and high pressure	Carboxylation
Methacrylate		Catalytic synthesis of CO_2 and propylene	Carboxylation

difficult to effectively support the formed fractures [172]. In addition, the high cost of capturing, pressurizing, transporting CO_2 and separating and repressurizing the backflow CO_2 are not negligible [174]. From this aspect, the feasibility of supercritical CO_2 fracturing technology depends on whether it can cost-effectively improve the production efficiency of crude oil or natural gas. Or, the increased production can compensate for the increased cost of replacing other fracturing technologies. This also requires field application and reliable yield verification.

3.2. Chemical utilization

In the chemical industry, CO_2 is an important raw material. Using CO_2 as a reactant, through appropriate chemical reactions, many chemicals such as urea, salicylic acid, dimethyl ether, organic acids, hydrocarbons, alcohols, amines, esters, and polyesters can be synthesized [175]. Table 5 summarizes several chemicals and synthesis methods converted from CO_2. As with physical utilization, chemical products synthesized by chemical utilization will eventually be oxidized on different time scales to CO_2 that will be emitted into the atmosphere. Chemical products generated by chemical conversion of CO_2 can be classified into three categories: bulk chemicals, organic fuels and polymer materials [176].

3.2.1. Synthetic bulk chemicals

The use of CO_2 to synthesize bulk chemicals is an effective way to enhance the economic value of CO_2. In traditional chemical industry applications, the most representative chemical products for CO_2 utilization are urea and salicylic acid, and the production of urea is the most important chemical utilization pathway for CO_2 [180].

3.2.1.1. Urea. The main methods of industrial production of urea are the reaction of ammonia and CO_2 [181]. The preparation process of urea consists of two reaction stages [182].

Synthesis of methylammonium by supercritical NH_3 and CO_2:

$$2NH_3+CO_2 \rightarrow NH_4COONH_2$$

Dehydration of methylammonium to form urea:

$$NH_4COONH_2 \rightarrow NH_2CONH_2+H_2O$$

The synthesis of urea consumes about 80% of the global ammonia output, and several process conditions for urea synthesis in this way are high temperature and high pressure [181,183]. Therefore, new catalytic and electroreduction technologies suitable for room temperature and atmospheric pressure conditions have become the exploration direction of many researchers.

Chen et al. [183] found that PdCu alloy nanoparticles on TiO_2 nanosheets could be used as an effective catalyst for simultaneous electrocatalytic coupling of N_2 and CO_2 to achieve green synthesis of urea. Under the action of catalyst, *N═N * spontaneously formed C-N bond with CO formed by reduction of CO_2 to promote urea synthesis. Urea yield and Faraday efficiency were 3.36 mmol h^{-1} g^{-1} and 8.92%, respectively. It may be a good research direction to synthesize urea with N_2 instead of NH_3. Yuan et al. [184] used InOOH nanoparticles to construct frustrated Lewis pairs (FLPs), which also achieved targeted adsorption and activation of N_2 and CO_2 molecules, effectively catalyzed the electrosynthesis of urea. Urea yield and Faraday efficiency were up to 6.85 mmol h^{-1} g^{-1} and 20.97%, respectively. Meng et al. [185] found that the reaction of CO_2 and nitrite ions could be catalyzed by ZnO (ZnO-V) porous nanosheets with self-loaded oxygen vacancy to realize urea electrosynthesis (Fig. 18). The Faraday efficiency of urea was 23.26%.

3.2.1.2. Salicylic acid. Salicylic acid can be synthesized by CO_2 and phenol at high temperatures and pressure [175]. Similar to urea synthesis, the salicylic acid synthesis process has poor conditions, long reaction time and large energy consumption. Therefore, it is of great significance to find a mild environment for salicylic acid synthesis.

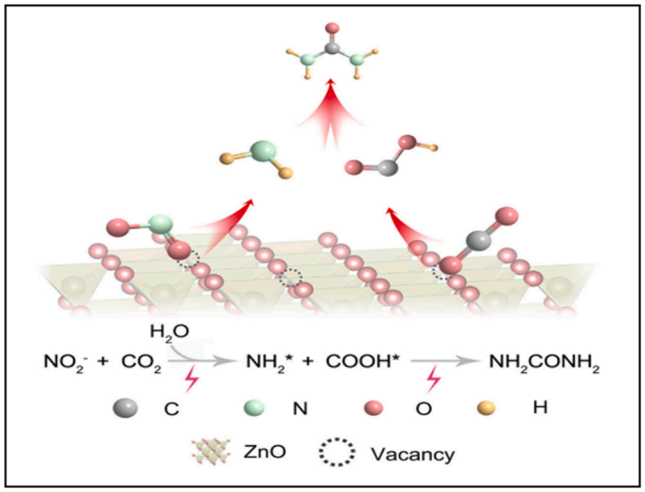


Fig. 18. Schematic diagram of the electrosynthesis of urea catalyzed by ZnO-V [185].

Iijima et al. [186] found that in the presence of Lewis acid catalyst, supercritical CO_2 and phenol can complete the synthesis of salicylic acid at moderate temperature. Adding 10 mmol of aluminum bromide under certain conditions, the phenol was almost completely converted to salicylic acid. Iijima et al. [187] also compared the catalytic effects of various alkali metals. It was found that potassium carbonate also had a good effect on the synthesis of salicylic acid from supercritical CO_2 and phenol. Under certain conditions, the conversion rate of phenol was 69.3%, and the yield and selectivity of salicylic acid could reach 69.3% and 99% respectively. Zhang et al. [188] also conducted the similar experiment. The catalytic performance of 11 alkali metal catalysts was compared, and the conclusion was consistent with that of Iijima et al., potassium carbonate showed the best catalytic effect.

3.2.2. Synthetic organic fuel

The captured CO_2 can be synthesized into organic fuels such as methane, methanol, formic acid and dimethyl ether through catalytic hydrogenation [1]. Unlike conventional fossil fuels, the 'green fuel' produced by CO_2 hydrogenation does not generate additional CO_2 emissions after use [189]. After the fuel is used, CO_2 is released back into the atmosphere, where it can be recaptured to synthesize the fuel, creating a carbon-neutral cycle (Fig. 19) [190].

3.2.2.1. Methane. Methane is an important domestic and industrial fuel and the main component of natural gas. When the ratio of H_2 to CO_2 in

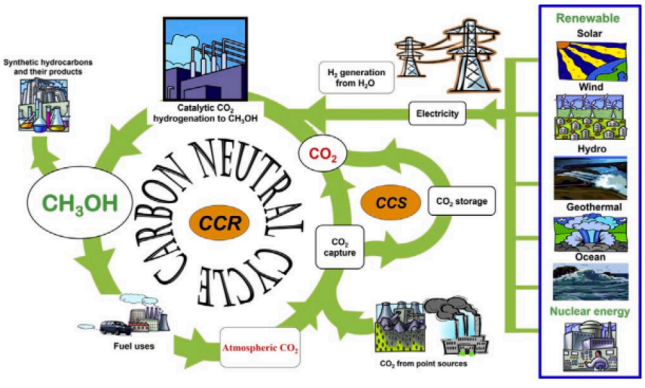


Fig. 19. Schematic diagram of carbon neutralization cycle [190].

CO_2 hydrogenation is equal to 4, results in CH_2 formation through the methanation reaction [191,192]. The CO_2 methanation reaction was proposed by the French chemist Paul Sabatier, so it is also called the Sabatier reaction [176]. The reaction formula is as follows [193]:

$$4\ H_2 + CO_2 \rightarrow CH_4 + 2\ H_2O$$

The reaction is exothermic and requires heating in the presence of catalysts [194]. At present, after a lot of explorations, the active components Co, Ni, Ru, etc. have been reported as efficient catalysts [195].

Liu et al. [196] loaded Co onto KIT-6 molecular sieve for catalytic CO_2 hydrogenation. It was found that the loading of Co affected the hydrogenation of CO_2 to CH_4. When the Co loading was 25 wt%, the CO_2 catalytic effect and CH_4 selectivity were the best, but the by-product CO was generated. Liu et al. [197] also investigated the effect of reduction temperature on the formation and catalytic effect of Co/KIT-6 catalysts. The results indicated that lower or higher temperature was not conducive to the reduction of Co^{2+}. When the reduction temperature was 400 °C, the Co/KIT-6 catalyst can be effectively prepared, and the catalytic activity of CO_2 hydrogenation and CH_4 selectivity were the best at this time, and the by-product CO was less. Chen et al. [198] used a facile mechanochemical ball-milling method to incorporate Ni into MgO/MgH_2 materials for thermo-catalytic CO_2 methanation. The results illustrated that mechanochemical ball milling enabled Ni and MgO to form a stable "Mg-Ni-O" structure, which promoted the hydrogenation reaction of CO_2. Zhou et al. [194] introduced $NaBH_4$ into the traditional Ru/TiO_2 catalyst, which changed the reaction pathway of CO_2 methanation. The Ru/TiO_2 catalyst with the addition of $NaBH_4$ improved the catalytic activity, CO_2 conversion and CH_4 generation rate of CO_2 hydrogenation.

Based on CO_2 hydrogenation, Martins et al. [199] proposed a new adsorption-conversion cycle process, which simultaneously realized the separation and utilization of CO_2. A parallel adsorption reactor used in this process not only separates the biogas well (CO_2/CH_4 mixture) but also reacts the separated CO_2 with the incoming H_2 to produce methane. This is a very favorable direction for the capture and utilization of CO_2, but the optimization of process operating conditions is the key to the further development of this research direction.

3.2.2.2. Methanol. The conversion of CO_2 to methanol is of particular importance because methanol has excellent combustion characteristics and is one of the important transportation fuels [1]. The catalytic hydrogenation of CO_2 can synthesize methanol, but it often leads to reverse water gas shift reaction (RWGS) [8].

Methanol synthesis reaction:

$$CO_2 + 3\ H_2 \rightarrow CH_3OH + H_2O$$

RWGS reaction:

$$CO_2 + H_2 \rightarrow CO + H_2O$$

The occurrence of RWGS reaction will consume part of H_2, resulting in lower methanol yield. At the same time, the two reactions will bring an increase in the production of water, thereby weakening the catalytic effect of the catalyst [8]. Therefore, the selection of catalyst is the key to the catalytic hydrogenation of CO_2 to methanol.

Currently, the most studied catalyst is the copper catalyst. However, Cu nanoparticles (NPS) can produce agglomeration effect under reaction conditions, resulting in the reverse water gas shift reaction catalyzed by the Cu surface, which decline the methanol yield [200]. Therefore, how to effectively limit the catalysis of RWGS reaction on Cu surface t has become the research direction of many scholars. Li et al. [201] explored the catalytic mechanism of Cu/ZrO_2 catalyst for CO_2 hydrogenation to methanol. It was found that ZrO_2 could interact with Cu due to its unique crystal structure, which improved the dispersion of Cu, stabilized Cu NPs in catalytic CO_2 hydrogenation, effectively inhibited the RWGS reaction, and promoted the generation of methanol. An et al. [200] improved strong metal-supported catalysts such as Cu/ZrO_2 and Cu/ZnO, and introduced MOFs to provide a novel Cu/ZnO_X @MOF catalyst for CO_2 hydrogenation. This catalyst utilized the porous structure of MOF to confine the agglomeration of Cu NPs, creating a unique opportunity for CO_2 hydrogenation to methanol. Qi et al. [202] also introduced MOFs and synthesized Cu-ZnO^{MOF}⊂Al_2O_3 catalyst by using a bimetallic MOF template, which made Cu NPs have higher dispersibility, thereby inhibiting the RWGS reaction and providing high activity and high selectivity for CO_2 hydrogenation to methanol.

3.2.2.3. Formic acid. Formic acid is considered a potential raw for low temperature fuel cells [1]. The reaction formula of CO_2 hydrogenation to synthesize formic acid is as follows:

$$CO_2 + H_2 \rightarrow HCOOH$$

At present, homogeneous catalysts are the most studied catalysts for the CO_2 hydrogenation to formic acid. Compared with the heterogeneous catalysts, the homogeneous catalytic systems have milder conditions and higher catalytic efficiency. The reaction mechanism is as follows: CO_2 interacts with the metal M-H bond to form complex intermediates; Subsequently, the M-H bond breaks, and the hydrogen atom is transferred to C to generate a C-H bond, forming a methoxy intermediate; Then the methoxy intermediate reacts with hydrogen to generate formic acid, which then reacts with alkali to convert to formate or ester, resulting in catalyst reduction [203]. This is the catalytic cycle of CO_2 hydrogenation to formic acid (Fig. 20).

Water-soluble catalysts are commonly used in systems for CO_2 hydrogenation to formic acid [204]. Zhao et al. [205] used $RhCl(mtppms)_3$ (mtppms=monosulfonated triphenylphosphine) as a catalyst to hydrogenate CO_2 in sodium formate solution to formic acid. The pressure ratio of sodium formate concentration to H_2 or CO_2 in the experiment affected the formation of formic acid. It was difficult to form formic acid without adding sodium formate to the solution. Therefore, it can be speculated that sodium formate promotes CO_2 hydrogenation in the presence of $RhCl(mtppms)_3$ catalyst. Coşkuner et al. [206] found that sodium borohydride ($NaBH_4$) in an aqueous solution could hydrogenate CO_2 in the presence of nickel-based catalysts. In the experiment, CO_2 was introduced into the NaBH4 solution. Under the action of a nickel-based catalyst, $NaBH_4$ was hydrolyzed and formed intermediate products such as formate with CO_2. These products were dissolved and heated in an alkaline aqueous solution to obtain solids such as formic acid and $NaB(OH)_4$. The catalytic cycle can be achieved if $NaB(OH)_4$ can be regenerated to form $NaBH_4$.

3.2.2.4. Dimethyl ether. Dimethyl ether (DME) is a good vehicle auxiliary fuel [204]. DME can be directly prepared by CO_2 hydrogenation, or be dehydrated from methanol produced by CO_2 hydrogenation [207]:

$$2CH_3OH \rightarrow CH_3OCH_3 + H_2O$$

$$2CO_2 + 6\ H_2 \rightarrow CH_3OCH_3 + 3\ H_2O$$

Since DME can be generated by methanol dehydration, the catalyst

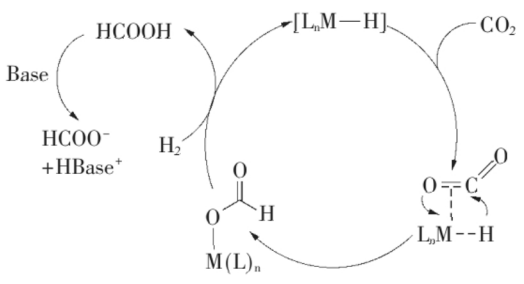


Fig. 20. Homogeneous catalytic reaction mechanism of CO_2 hydrogenation to formic acid [203].

used for CO_2 hydrogenation to dimethyl ether is generally a composite catalyst composed of methanol synthesis catalysts and dehydration catalysts such as molecular sieve [204]. Sun et al. [208] used Pd to modify the CuO–ZnO–Al_2O_3–ZrO_2/HZSM-5 composite catalyst for the synthesis of DME via hydrogenating CO_2. The results showed that when 0.5 wt% Pd was added to the catalyst, the CO_2 conversion was 18.67%, and the selectivity of DME reached 73.56%. Ren et al. [209] modified the CuO/ZnO/Al_2O_3(CZA)catalyst for methanol synthesis by adding different contents of Al and Zr before forming a composite catalyst to obtain CuO/ZnO/ZrO_2/Al_2O_3 (CZZA) catalyst. Under the action of the CZZA and HZSM-5 composite catalyst, the synthetic stability of DME was significantly improved, and the maximum yield was 18.3%. Fang et al. [210] obtained a novel composite catalyst Cu-ZnO-ZrO_2/HZSM-5 (CZZ/HZSM-5) by in situ growth of HZSM-5 on Cu-ZnO-ZrO_2 (CZZ) nanoparticles through the action of aluminum. The catalyst expressed excellent catalytic performance. Under certain reaction conditions, the CO_2 conversion was 34.1%, the selectivity of DME was 67.7% and the yield of methanol + DME was 25.8%. Compared with the traditional physical mixing method, in situ growth can shorten the distance between CZZ and HZSM-5, promote methanol dehydration, and improve product yield.

3.2.3. Synthetic polymer materials

The synthesis of CO_2 polymers with the economic value from CO_2 is an important research direction in the field of CO_2 chemical utilization [211]. From the perspective of molecular structure, CO_2 has the conditions for copolycondensation and addition copolymerization to form polymer materials [180]. Combined with decades of research work at home and abroad, the synthesis of polycarbonate (PC) and polyurethane (PU) from CO_2 are considered to be the two most practical methods (Fig. 21) [211].

3.2.3.1. Polycarbonate. Polycarbonate (PC) is a kind of thermoplastic polymer containing carbonate units [212,213]. PC synthesized with CO_2 as comonomer has biodegradability and is a potential sustainable plastic [214]. At present, the preparation of aliphatic polycarbonate (PPC) by copolymerization of CO_2 and epoxide is the best and most common way (Fig. 22) [215].

Since Inoue first discovered the copolymerization of CO_2 and propylene oxide in the presence of catalysts in 1969, the development of new catalysts and catalytic systems has become the core of this research. Wei et al. [216] reported the use of bimetallic cyanide complexes (Zn-Co (III)DMCC) to catalyze the formation of polycarbonate from epichlorohydrin and CO_2. A novel bimetallic nickel complex prepared by Liu et al. [217] also effectively catalyzed the copolymerization of CO_2 and epoxides. Under the catalysis of this complex, CO_2 could be combined with cyclohexene oxide to synthesize CO_2-based poly(cyclohexene carbonate)s (PCHCs) with carbonate linkages greater than 99%, and polycarbonate with vinyl functional groups could also be synthesized with

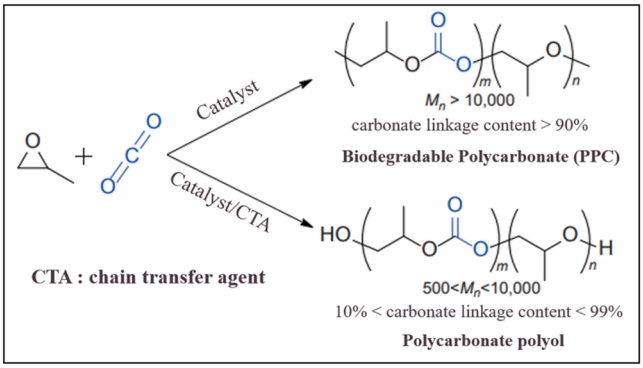


Fig. 21. Synthetic routes of two CO_2 polymers [211].

4-vinyl-1,2-cyclohexene oxide. Cui et al. [218] used catalyst SalenCoCl and cocatalyst bis(triphenylphosphine)iminium chloride (PPNCl) to realize the copolymerization of soybean oil-based terminal epichlorohydrin and CO_2, and developed a sustainable method for the synthesis of biological polycarbonate. The yield of biological polycarbonate reached 83.3%.

3.2.3.2. Polyurethane. Polyurethane (PU) is a polymer material commonly used in the manufacture of various foams and plastic sponges. PU appeared in the 1930 s. After decades of technological development, it has been widely used in various fields. Isocyanates are used in traditional PU synthesis methods, but are highly toxic [219]. Therefore, finding a harmless PU synthesis method has become the research direction of many scholars at home and abroad. Replacing isocyanate with CO_2 can not only alleviate the problem of CO_2 emissions, but also realize the resource utilization of CO_2, which is undoubtedly a new breakthrough on the road of PU synthesis.

Ying et al. [220] used CO_2 as a carbonyl source to replace isocyanate and amino-modified polysiloxane to prepare novel polyurethane. The results showed that the synthesized polyurethane remained stable and hydrophobic under the condition of lower than 150 ℃. In addition, polyurethane with high urea bond content had a better absorption effect on some organic solvents. Jiang et al. [221] synthesized polyurethane thermoplastics materials by polycondensation of CO_2 with mixed binary diamines. The results revealed that, compared with the polyurethane synthesized by CO_2 and single diamine, the polyurethane synthesized by the mixture of CO_2 and diamine showed higher performance, and the mechanical properties of the polyurethane were better than those of polycarbonate and polyamide. Zhao et al. [222] reported a solvent-free and catalyst-free route for the indirect synthesis of polyurethane from CO_2. First, CO_2 storage materials (CO_2SMs) were prepared by absorbing CO_2 with diamine + glycol system at room temperature, and then polyurethane was synthesized by intramolecular dehydration of the storage material.

3.3. Biological utilization

CO_2 bio-utilization technology refers to the process of designing new artificial photosynthetic systems and pathways by simulating the photosynthesis of plants or microorganisms in nature, so as to efficiently convert CO_2 into different high-value products [223]. The current research on CO_2 bio-utilization mainly focuses on the conversion of plant aerosol fertilizers and microalgae carbon sequestration into biofuels.

3.3.1. Plant gas fertilizer

Using CO_2 in plant photosynthesis is the most direct way of biological utilization, with inherent effectiveness and sustainability. At the same time, CO_2 is an indispensable gas in the process of plant growth. Under closed conditions, a certain amount of CO_2 is introduced into the greenhouse, which is conducive to plant photosynthesis and promotes plant growth. Utilizing CO_2 as a plant gas fertilizer can increase plant yield, which is simple to operate and low in cost. The amount of CO_2 released from the greenhouse depends on the initial concentration. Within a certain range, the higher the CO_2 concentration, the better the growth effect of plants [224–226]. At present, hanging plant gas fertilizer is mainly used in the market, which is convenient to operate and has remarkable economic benefits. It is an indispensable means for the high yield and income increase of modern greenhouse crops.

3.3.2. Carbon sequestration by microalgae to produce biofuels

Among CO_2 utilization pathways, the use of CO_2 to cultivate microalgae and convert them into biofuels is considered an attractive option [227,228]. Compared with the utilization method requiring high-purity CO_2 injection, microalgae can directly absorb CO_2 in flue gas

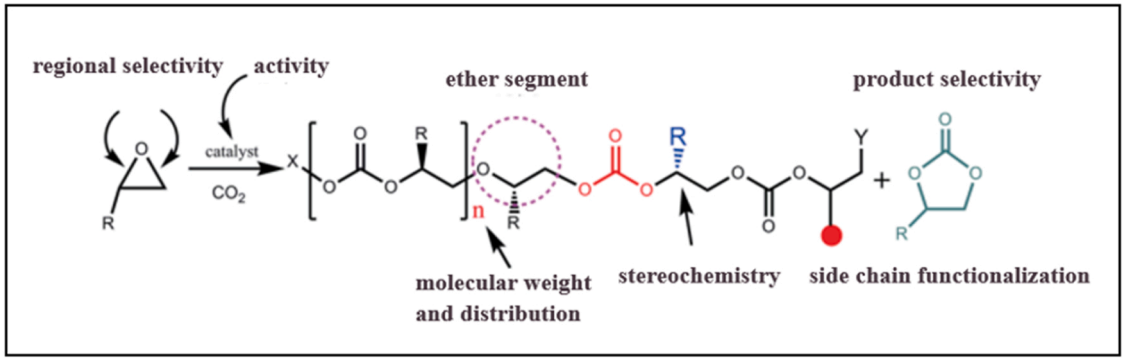


Fig. 22. Copolymerization of CO_2 and epoxide [211].

to complete the carbon sequestration process [229]. The process of microalgae carbon sequestration into biofuel is shown in Fig. 23. The microalgal biomass obtained through this process has high water content. Therefore, drying wet microalgal biomass is a crucial step before conversion into biofuels [230]. At the same time, the high energy consumption in the drying process is the main problem hindering the development of microalgae carbon sequestration technology to synthesize biofuels [1].

Microalgae can obtain different biofuels through three different conversion processes: chemical, biochemical and thermochemical [1] (Fig. 24). The chemical conversion of microalgae is mainly through transesterification to produce biodiesel [231]. This conversion process usually requires the addition of excess ethanol to promote the formation of light components in the biodiesel structure. The obtained biodiesel is non-toxic, degradable and renewable [231]. Biochemical transformation includes two types of anaerobic digestion and fermentation. Pretreatment of microalgal biomass is usually required for the conversion into methane, hydrogen and various alcohol products [232,233]. Thermochemical conversion is a promising conversion technology, including pyrolysis, hydrothermal liquefaction (HTL), and gasification [233]. Among them, the transformation process of HTL does not need an energy-intensive drying process, and is a treatment method that directly acts on wet microalgal biomass [234]. The resulting products are mainly bio-oil or bio-crude [235]. HTL is very attractive for the production of liquid fuels from wet biomass, and is expected to be a key technology to solve the process bottleneck problem of excessive drying energy consumption.

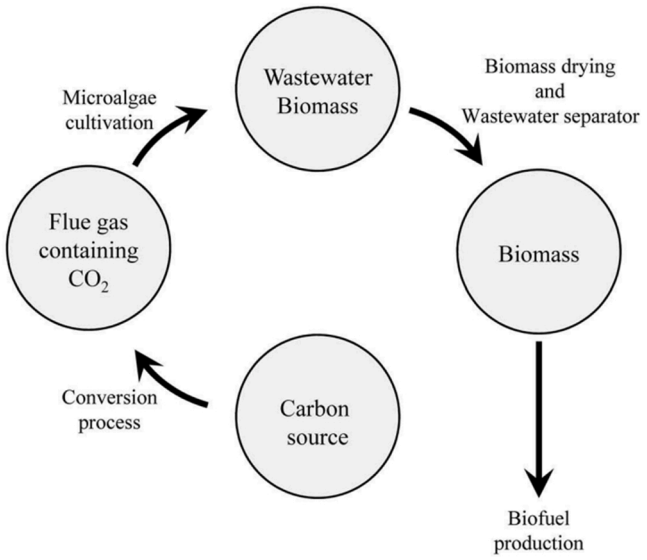


Fig. 23. Process of carbon sequestration and conversion of microalgae into biofuels [230].

The use of microalgae for carbon sequestration to produce biofuel has obvious advantages. Although the storage of CO_2 storage is reduced, the cultivation of microalgae cannot compete with the food market [231]. From a technical point of view, the use of CO_2 to cultivate microalgae and convert them into biofuels has great potential for production. However, the economics of this utilization approach are complex, and further research or technical improvements are required to support economic feasibility.

3.4. *Mineralization utilization*

CO_2 can form various carbonate precipitates through the mineralization process and be effectively utilized [236]. The different mineralization reaction processes can be divided into direct mineralization and indirect mineralization [237]. Direct mineralization refers to imitating the weathering process of CO_2 minerals in nature, so that the mineralized raw materials react directly with CO_2 [238]. At room temperature and pressure, this reaction proceeds very slowly [239]. Enhanced reaction conditions, such as increasing temperature and pressure, are often required to promote mineralization. In addition, the mineralization cost of natural ore is too high, and it is difficult to convert the mineralized products into valuable products for sale [1,240]. Indirect mineralization (Fig. 25) refers to the process of converting mineralized raw materials into intermediate products with a certain chemical additive, and then reacting with CO_2 [237]. This process improve the reaction rate of CO_2 mineralization, and can obtain different high value-added products [1]. However, the use of chemical additives increases energy consumption and associated costs [237].

The raw materials that can be used for CO_2 mineral storage are natural silicate ores and alkaline oxides in industrial solid residues, such as CaO and MgO, which can react with CO_2 to form stable inorganic carbonates [236,241]. This carbonate stores CO_2 in an extremely stable form, which provides a safe and effective way for permanently sequestering CO_2 [238]. It is worth noting that the premise of permanent storage is that the carbonate is not used for some activities that can release CO_2 from the salt. Inorganic products obtained by CO_2 mineralization of industrial solid wastes rich in calcium and magnesium (such as blast furnace slag, steel slag, waste gypsum, and fly ash) are a potential building material [238]. While realizing CO_2 emission reduction, carbonated products with economic benefits can be obtained. Therefore, CO_2 mineralization has a certain potential for industrial utilization.

Natural ore has strong CO_2 capture capacity and rich ore reserves [1]. However, some problems inherent in natural ores, such as low mineralization reaction rate of and poor conversion ability, which limit their large-scale application [242]. Therefore, the use of natural ores to sequester carbon may have little effect on the rationale for addressing today's rising carbon stocks. Compared with natural silicate ores, CO_2 mineralization of industrial solid residues has a faster reaction rate, higher conversion efficiency and lower energy input [238]. At the same

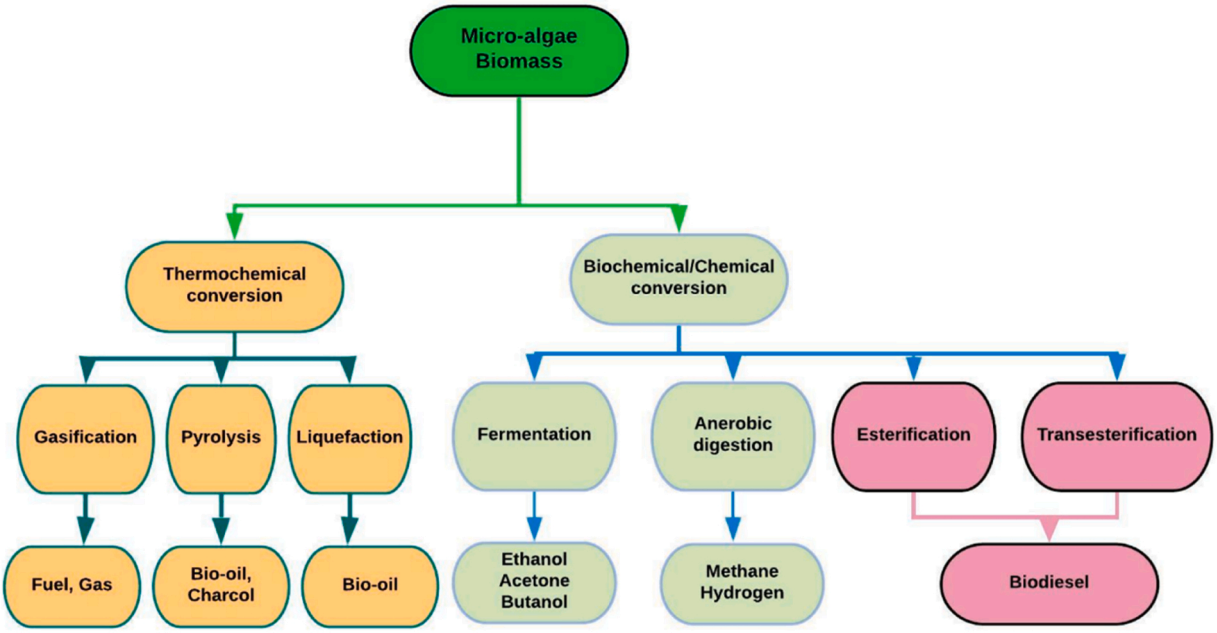


Fig. 24. Different conversion processes of microalgae biomass for biofuel production [233].

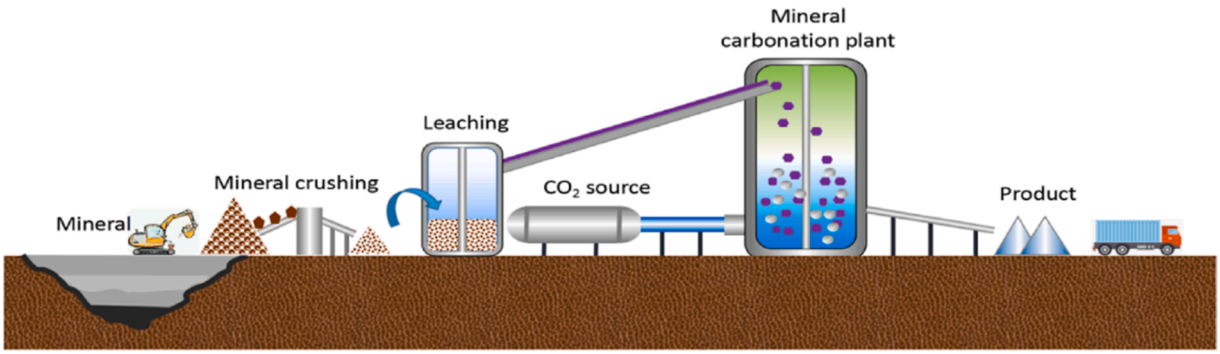


Fig. 25. Schematic diagram of CO_2 indirect mineral carbonization [238].

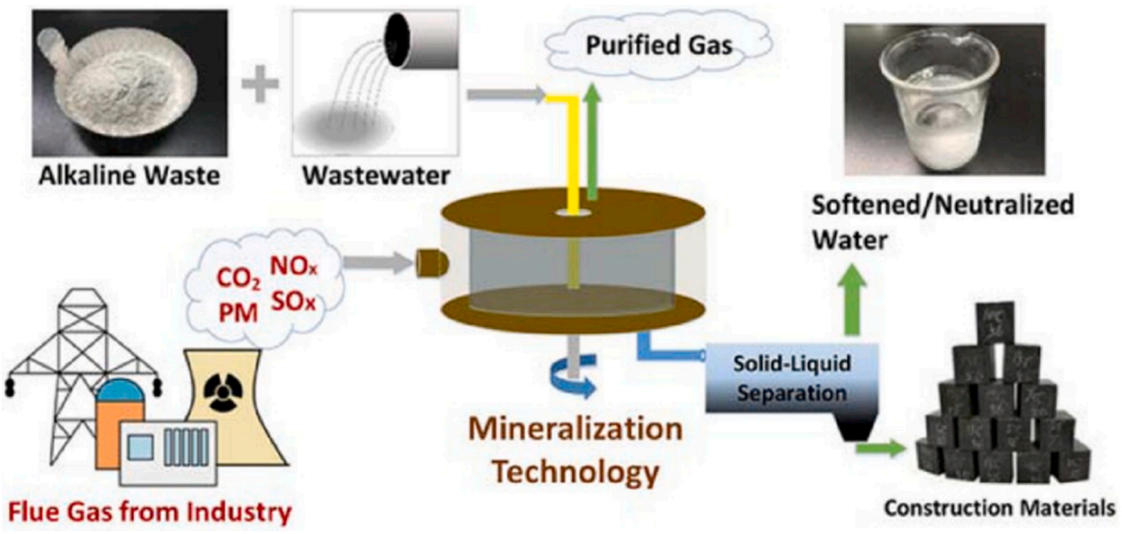


Fig. 26. The combination process of CO_2 mineralization utilization and industrial air pollution control [236].

time, it is relatively convenient to reclaim industrial solid residues. A large amount of solid waste, such as coal ash, coal gangue, carbide slag and steel slag, are produced in various factories every year [243]. These solid residues severely damage soil, air and water resources. If these solid residues containing alkaline oxides are filled into the goaf after mining, it is expected to form an effective carbon storage space [244]. The collapse of the overlying strata in the goaf is avoided while CO_2 is mineralized. At present, many studies have proved that it is effective and feasible to modify industrial wastes and fill them into goaf through grouting [245–247].

In addition, CO_2 mineralization of alkaline solid waste is a good option for controlling industrial air pollutant emissions (Fig. 26). Pei et al., Chen et al. [248,249] used fly ash to verify the feasibility of removing industrial air pollutants by high-gravity mineralization process. The results showed that the removal efficiency of CO_2, SO_2 and nitrogen oxides reached more than 95%, and the removal efficiency of the particulate matter reached 83%. What is more, the reacted fly ash could be used as supplementary gel materials for concrete or cement mortar. Compared with the fly ash before the reaction, the cement prepared with carbonated fly ash had higher compressive strength and durability.

CO_2 mineralization utilization technology is considered as one of the efficient CO_2 conversion schemes. CO_2 emission reduction is achieved while reducing industrial solid waste. At the same time, the mineralized waste can be sold as products, bringing additional income in certain industrial fields and improving the economy of CO_2 and solid waste resource utilization. However, the current research on CO_2 mineralization utilization and the future industrial deployment still face great challenges. The reaction system and mineralization process need to be further optimized to reduce the high cost and energy loss caused by raw material grinding, stirring, heating, etc. [236]. The mineralization reaction is generally an exothermic reaction. Therefore, it is feasible to recover energy from the exothermic reaction to compensate for the energy consumption of CO_2 mineralization [237]. In addition, the effective combination of CO_2 mineralization with other aspects, such as the filling of mineral gobs and the emission control of industrial pollutants, may be the main research direction of CO_2 mineralization utilization in the future.

4. Frontier technologies of CO2 resource utilization

This part reviews the current frontier reports on CO_2 synthesis, and expands new research directions for the realization of CO_2 resource utilization.

4.1. Photoelectric drive to achieve simultaneous CO2 and H2S conversion

Ma et al. [250] reported a solar-driven electrochemical method for simultaneous conversion of CO_2 and H_2S, which provided a new idea for CO_2 conversion and natural gas purification. In this strategy, non-precious-metal (graphene-encapsulated zinc oxide) was used as the cathode catalyst to reduce CO_2, and graphene was used as the anode catalyst to oxidize EDTA-Fe^{2+}. EDTA-Fe^{3+} produced by oxidation reacted with H_2S to generate elemental sulfur and protons. Protons were transferred to the cathodic compartment for electrochemical reduction of CO_2 to CO (Fig. 27). This process achieved the simultaneous CO_2 and H_2S conversion.

4.2. Biotransformation of CO2 - efficient synthesis of fatty acids

Li et al. [251] constructed an effective pathway from CO_2 to organic biosynthesis based on Ralstonia eutropha. In this work, several pathways of Ralstonia eutropha H16 were modulated by manipulating heterologous and endogenous genes related to fatty acid synthesis. The modified strain B2 (PCT, PFP) was fermented with fructose as the sole carbon source to produce 124.48 mg/g free fatty acid, which was 4

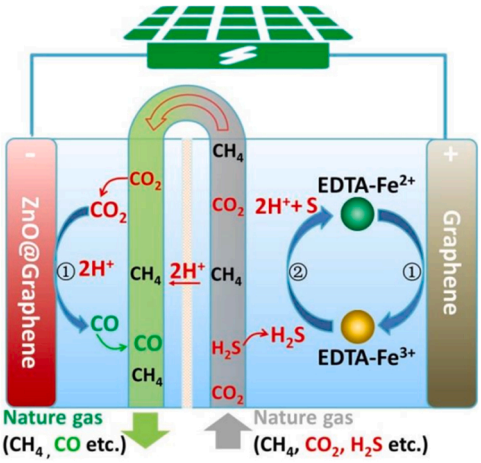


Fig. 27. Co-conversion process of CO_2 and H_2S [250].

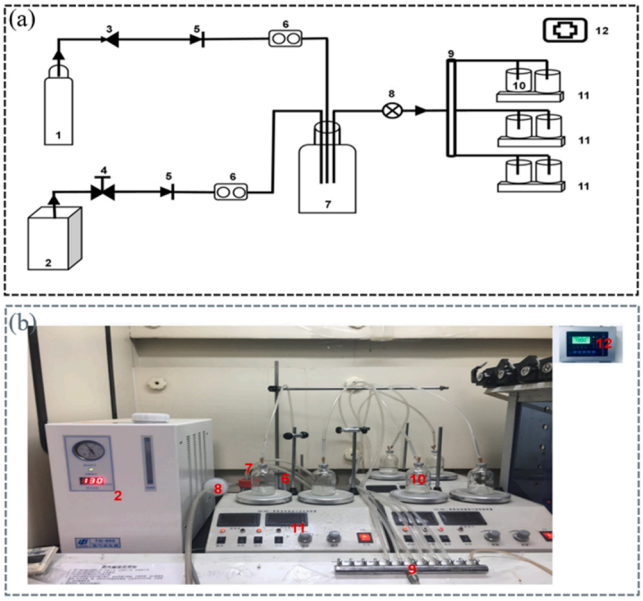


Fig. 28. (a) Design drawing of autotrophic fermentation system (b)Actual drawing of autotrophic fermentation system [251].

times higher than that of the control strain H16. The research team also developed an autotrophic fermentation system using H_2, CO_2 and O_2 as substrates (Fig. 28) to continuously produce of fatty acids. In this system, B2 (PCT, PFP) could be fermented to produce 60.64 mg/g of free fatty acids every 48 h. During the fermentation process, the ratio of H_2 to O_2 was kept at 7:1, and a gas detection device was equipped to ensure the safety of the experiment. This study laid the foundation for the microbial fixation of CO_2.

4.3. Bifunctional catalytic system for CO2 adsorption and conversion

Liu et al. [252] developed a novel periodic mesoporous organosilicon catalytic system modified with a squaramide-derived framework (PMO-SAF) (Fig. 29). At the same time of CO_2 capture, CO_2 could be catalytically converted to cyclic carbonates in a mild, solvent-free and metal-free environment. This method avoided the use of volatile solvents and toxic metal ions. In addition, PMO-SAF had the structural characteristics of high specific surface area, Lewis base and double hydrogen bond donor co-modification, which provided favorable conditions for CO_2 adsorption and conversion. This special design realized

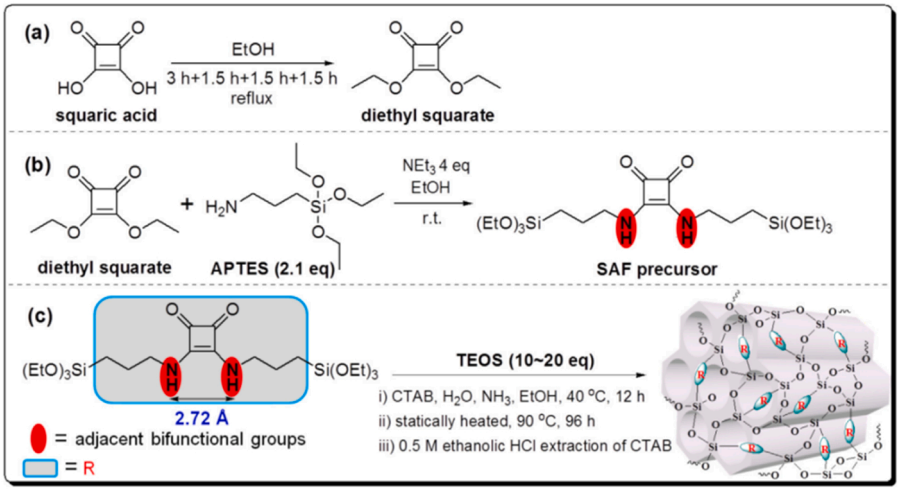


Fig. 29. Preparation scheme of PMO-SAF [252].

both capture and utilization of CO_2, providing new inspiration for the development of new catalytic materials.

4.4. *Efficient synthesis of artificial starch by CO2*

Cai et al. [253] first reported the complete synthesis process from CO_2 to starch molecules in the journal Science (Fig. 30). In this study, an 11-step core reaction was designed and constructed from scratch through modular assembly. The synthesis of artificial starch could be realized only by the chemical reaction between CO_2 and H_2. According to the detection, the structure of synthetic starch was the same as that of natural starch, and the synthetic rate of artificial starch obtained by this approach was about 8.5 times that of corn starch. This work constructed a starch synthesis route that does not exist in nature, and the synthesis

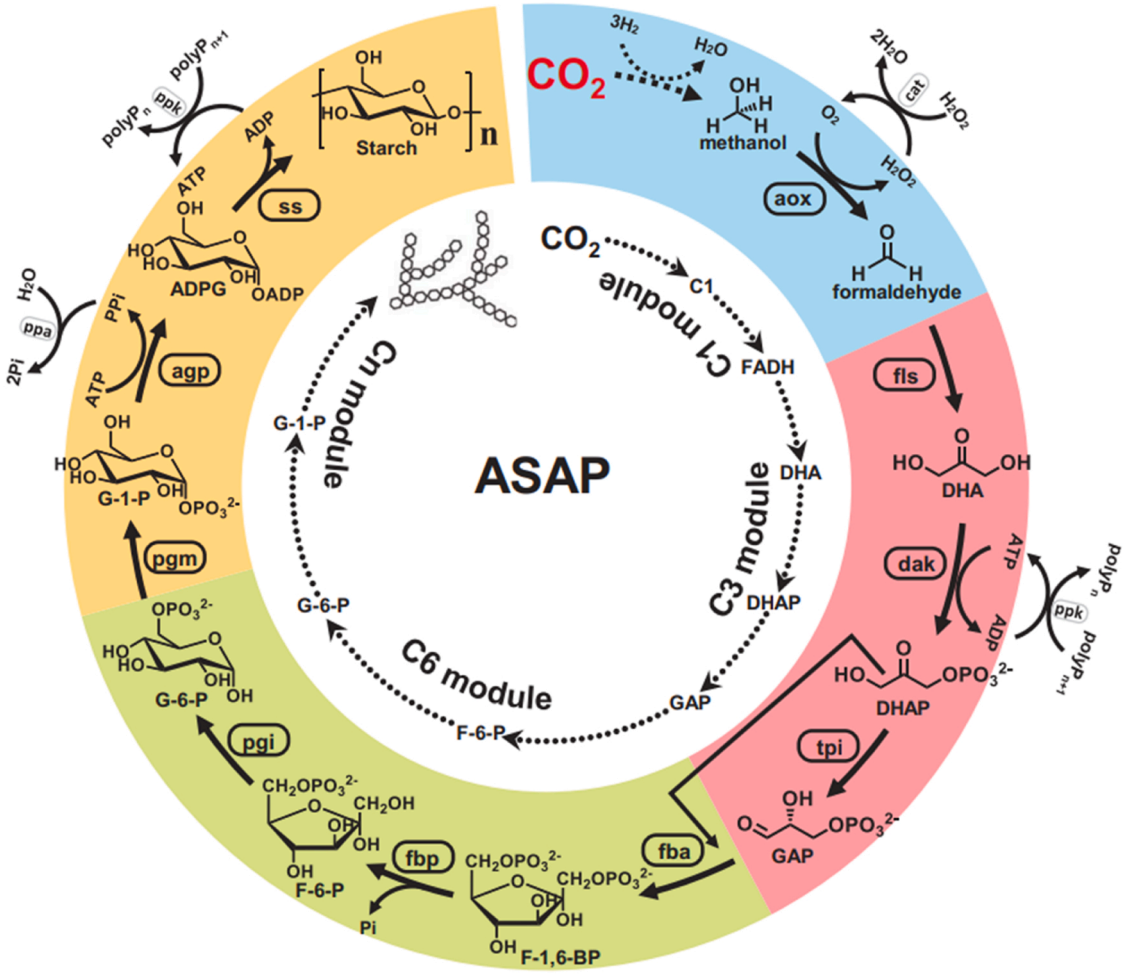


Fig. 30. Design and modular assembly of the artificial starch anabolic pathway (ASAP) [253].

effect was superior to natural biological processes, which opened up a new way for CO_2 to synthesize starch.

4.5. CO2 catalytic hydrogenation to synthesize gasoline

As early as 2017, Wei et al. [254] reported a multifunctional composite catalyst Na-Fe_3O_4/HZSM-5 that can directly convert CO_2 and H_2 into high octane gasoline in the journal Nature Communications. The catalyst had low methane selectivity (4%), and the selectivity of gasoline distillate hydrocarbon (C5-C11) in hydrocarbon products reached 78%. The reaction process of CO_2 hydrogenation to gasoline is shown in Fig. 31.

In March 2022, the world's first thousand-ton CO_2 hydrogenation-to-gasoline equipment jointly developed by Wei and other research teams and Zhuhai Fortune Energy Technology Co., Ltd. successfully produced standard clean gasoline products. This technology enabled the conversion rate of CO_2 and H_2 to reach 95%, which provided a new solution to the global greenhouse effect and oil shortage. However, the extraction of high-purity CO_2 and the production of hydrogen energy not only required high-cost investment, but also consumed a large amount of energy. Therefore, seeking more stable and cheaper CO_2 and H_2 energy sources will be the key to the next development of this technology.

5. Conclusion

The capture technologies and utilization directions for CO_2 are discussed. Various capture technologies have been comprehensively introduced, including chemical absorption, solid-phase porous materials adsorption, membrane separation, cryogenic separation method, and emerging methods such as hydrate method and microbial methods. The CO_2 utilization technologies are systematically discussed from the aspects of physical utilization, chemical utilization, biological utilization and mineralization utilization. More importantly, as the current research hotspot, the cutting-edge technologies of CO_2 resource utilization has been reported.

(1) Chemical absorption method has the largest capture capacity, and is most suitable for industrial application. But its energy consumption is also the largest. For the corrosive absorbents, the operating costs will further increase. Solving the energy consumption problem will be an important breakthrough on the research road of CO_2 capture by absorption method.
(2) For the solid-phase porous materials adsorption technology, the energy consumption is low, but the adsorption capacity is limited. Therefore, it is usually adopted as the terminal treatment for industrial CO_2 capture. MOFs and COFs materials show strong application potential due to their structural designability. The biggest problem for MOFs and COFs at present is the relatively high cost. In the future, the key to industrial production is to simplify the synthesis process and reduce the production cost.
(3) The low-temperature separation of CO_2 has the highest CO_2 purity is, but the process is complicated and the cost is high. This technology has not been used on a large scale in the industry at present.
(4) The membrane separation method, hydrate method and microbial method are all in line with the concept of green CO_2 capture, with low energy consumption. Among them, the membrane separation method is the most potential new capture technology. It is necessary to further improve the service life, permeability and selectivity of membrane, so as to realize large-scale applications.
(5) In the physical utilization of CO_2, CO_2 flooding for enhancing oil recovery is a promising way of CO_2 utilization and storage, which can bring considerable income and provide conditions for further CO_2 capture and storage.

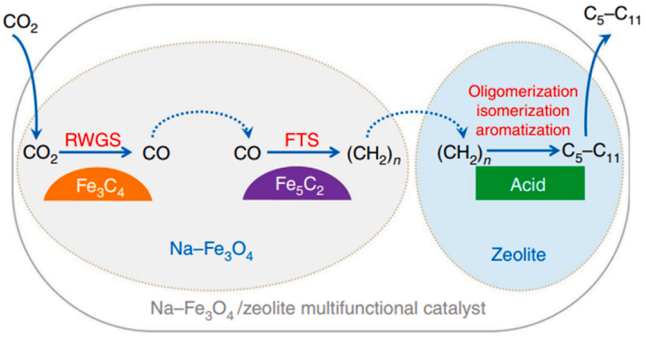


Fig. 31. Reaction scheme of CO_2 hydrogenation to gasoline [254].

(6) It is an effective and feasible way to convert CO_2 into organic fuels and chemical products. Its valuable products can offset the cost for carbon capture and conversion. How to further improve the utilization rate of CO_2 in a milder environment is the key to the further development of this technology. The catalytic conversion of CO_2 by photocatalysis and electroreduction technologies can improve the process conditions, which is a research direction with great potential to reduce the cost of carbon capture and utilization.
(7) Microalgae carbon sequestration into biofuel is an attractive pathway for CO_2 utilization. The energy consumption of microalgal biomass drying is the main problem to be solved, and hydrothermal liquefaction is expected to be the key technology of microalgal conversion process.
(8) The CO_2 mineralization and utilization technology realizes the efficient conversion of CO_2, and the mineralized products can be sold or used as high value-added products. The current research focus is to collect and utilize the heat generated by the mineralization reaction to neutralize the cost of energy consumption.
(9) The simultaneous conversion of CO_2 and other harmful gases, the introduction of metabolic engineering, and the novel catalytic system that enables simultaneous CO_2 adsorption and conversion should be the main research directions in the future; CO_2 hydrogenation to produce starch and gasoline is a major breakthrough in the field of food and energy. These synthetic directions achieve high-value utilization of CO_2 in different ways. The next step is to explore more stable CO_2 and hydrogen energy sources to support the sustainable development of synthesis technology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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